Why Temporal Abstraction Can Fail in Imagination-Based RL: A Diagnostic Analysis of Variable-Rate DreamerV3

Anonymous Authors[1]

Abstract

World-model reinforcement learning is increasingly deployed for robot control, where temporal abstraction enables planning across heterogeneous time scales. We find that Δt-conditioning, effective in model-predictive control (MPC), cannot transfer to imagination-based actor-critic training: every variable-rate DreamerV3 configuration degrades to 14–19% of baseline. Factorial diagnostics on two continuous-control tasks identify trained temporal embedding with multi-rate data exposure as the strongest predictor of this failure—*conditioning-induced value overestimation* (CIVO; n=36, AUROC=1.000, F1=1.000). The RSSM develops a reward prediction gap at Δt>1 that the critic internalizes as optimistic values; the contrast with MPC identifies feedback architecture as a hypothesized structural factor. No tested intervention across four levels (gradient, output, representation, architecture) reliably eliminates CIVO: dual critics converge in pure imagination, so Double Q pessimism is ineffective, though min + increased capacity enables recovery.

1. Introduction

World-model reinforcement learning enables sample-efficient learning directly on physical hardware (Wu et al., 2023; Hansen et al., 2024). Real robotic tasks demand temporal abstraction: manipulation requires fast reactive grasping alongside slow approach planning, locomotion involves gait phases at different frequencies, and sim-to-real transfer must handle dt mismatches between simulation and hardware. TAWM (Nhu et al., 2025) shows that conditioning a world model on step size dt yields consistent gains within TD-MPC2's MPC framework (Hansen et al., 2024). Can the same dt-conditioning bring temporal abstraction to imagination-based actor-critic methods like DreamerV3 (Hafner et al., 2025a), where policies are trained entirely in imagined rollouts?

We find that it cannot. Every variable-rate configuration degrades performance catastrophically (14–19% of baseline). Through a factorial analysis on Walker Walk (Tassa et al., 2018), we isolate the combination of a *trained* dt-embedding and multi-rate data exposure as the strongest predictor. This combination enables environment-dependent pathologies—a $1.63\times$ reward prediction gap in Walker Walk, or dynamics-model prior drift without measurable reward bias in Cheetah Run—which the critic internalizes as optimistic value estimates, a failure mode we term *conditioning-induced value overestimation* (CIVO), analogous to delusional bias (Lu et al., 2018) but model-induced rather than policy-class-induced. A normalized value-to-eval (NVE) threshold of $\geq 2.0\times$ achieves perfect classification (n=36; AUROC=1.000, F1=1.000; Appendix B).

A four-level intervention hierarchy (gradient, output, representation, architecture) shows that no tested intervention reliably eliminates CIVO; in pure-imagination training, dual critics converge ($\rho > 0.999$), so Double Q pessimism is ineffective. These findings are directly relevant to robotics: as world-model RL moves toward real systems with variable-rate sensing, understanding when temporal conditioning fails is essential for safe deployment.

Contributions.

1. **Failure mode identification.** We identify CIVO as specific to imagination-based actor-critic with temporal conditioning and isolate trained dt-embedding + multi-rate exposure as jointly sufficient (5/7 multi-joint locomotion seeds, 0/16 frozen or removed controls; AUROC=1.000, F1=1.000). Cross-environment validation (six environments, three task types) confirms specificity to multi-joint locomotion (§5).
2. **Architectural boundary.** dt-conditioning succeeds in TD-MPC2's MPC re-planning but fails under imagination-based bootstrapping, where model errors compound into biased value targets with no real-data

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

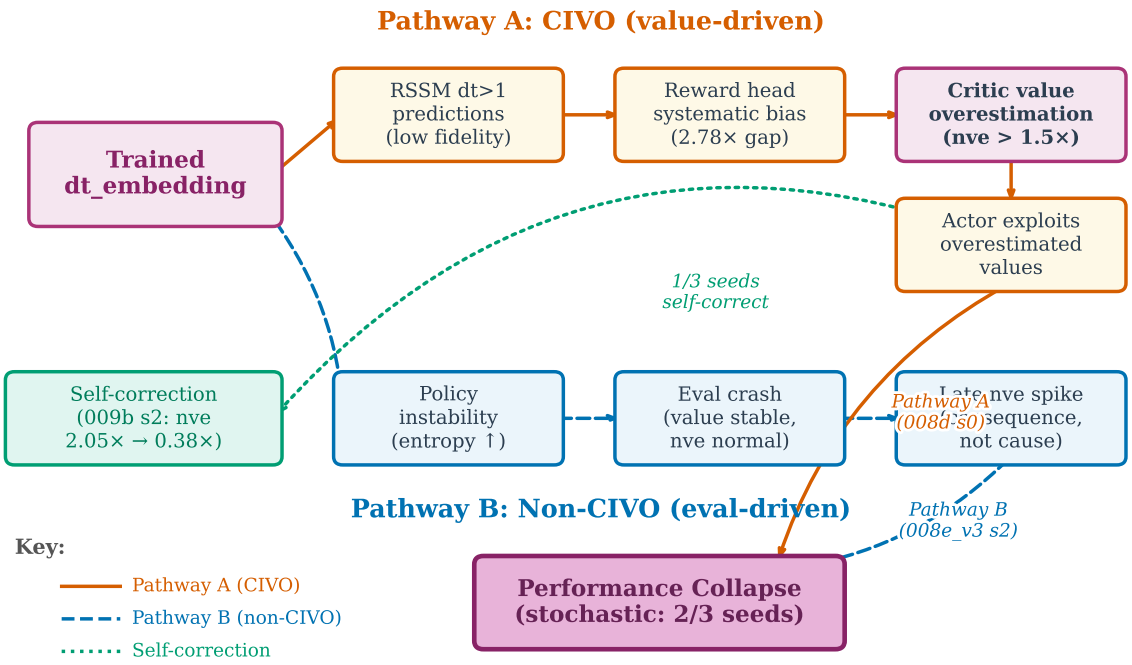


Figure 1. CIVO mechanism. The trained dt-embedding enables asymmetric reward prediction fidelity, initiating a feedback loop: biased RSSM rewards → optimistic critic values → actor exploitation → reinforced state-distribution skew.

anchor.

3. **Four-level intervention hierarchy.** No intervention reliably eliminates CIVO across environments; gradient magnitude does not predict causal importance. Dual critics converge in pure imagination ($\rho > 0.999$), so Double Q pessimism vanishes; the min operation enables recovery through a separate mechanism.
4. **Preliminary diagnostic.** A proof-of-concept online detector (distinct from the NVE threshold classifier) achieves TPR 95% CI $[47.8\%, 100\%]$ (5/5 positives), FPR 22% (2/9), with predictive warning in 2/5 cases (n=14 runs, three environments; §4.7).

2. Related Work

Temporal abstraction in RL. The options framework (Sutton et al., 1999), option-critic (Bacon et al., 2017), HAM (Parr & Russell, 1998), and MAXQ (Dietterich, 2000) formalize temporal abstraction through hierarchical policies. Semi-MDP theory (Bradtke & Duff, 1995) provides the mathematical foundation. Action-repetition methods (Sharma et al., 2017; Biedenkapp et al., 2021; Park et al., 2021) achieve simpler temporal abstraction via real transitions; Wang & Beltrame (2024) learn elastic time steps that adapt step size to state dynamics. The Dreamer family (Hafner et al., 2020; 2021; 2025a;b) trains policies in imagination at fixed dt=1; hierarchical variants (Hafner et al., 2022; Gumbsch et al., 2024; Mattes et al., 2024; Li et al., 2025) use subgoals or separate dynamics per scale, avoiding flat dt-conditioning. TAWM (Nhu et al., 2025) conditions a world model on dt within TD-MPC2 (Hansen et al., 2024) and achieves consistent gains; the critical distinction is that MPC re-plans from real states, so model errors do not accumulate into a critic (§5).

Value overestimation and model fidelity. Maximization bias (Thrun & Schwartz, 1993) has been addressed by Double Q-learning (van Hasselt et al., 2016), TD3 (Fujimoto et al., 2018), and ensemble methods (Chen et al., 2021); in model-based settings, MBPO (Janner et al., 2019) and MOPO (Yu et al., 2020) bound rollout error, while MOReL (Kidambi et al., 2020) and COMBO (Yu et al., 2021) penalize model uncertainty. Lu et al. (2018) identify delusional bias from policy-class constraints. CIVO is mechanistically distinct: the amplifier is the world model's co-adaptation with a learned temporal embedding, producing systematic reward-prediction bias at $\mathrm{dt} > 1$. Multi-timestep world models (Benechehab et al., 2023) learn predictions at variable horizons but do not examine the value overestimation pathology we isolate. Compounding error in multi-step rollouts (Lambert et al., 2020; Asadi et al., 2019) is a related but distinct phenomenon: if CIVO were generic compounding error, untrained dt-embeddings at dt=2 should also produce overestimation, yet they do not (NVE<0.6× for 008a/008c vs. 4.12× for 008d). Signed reward error analysis reinforces this distinction: dt=2

rewards are *under*predicted (mean SE$=-0.599$), which rules out observation-level overprediction and localizes CIVO to imagination-level prior drift (§5). Decision-aware model learning (Hamrick et al., 2021) optimizes for planning performance but does not address dt-conditioned co-adaptation. Our TCN intervention relates to training-with-noise (Bishop, 1995); TCN's environment-specific dose sensitivity (§4.5) suggests CIVO is a structural vulnerability, not a regularization deficiency. Our factorial diagnostic follows Henderson et al. (2018).

3. Experimental Setup

3.1. Base Architecture

All experiments build on DreamerV3 (Hafner et al., 2025a), which learns a recurrent state-space model (RSSM) with discrete latent states (32 categories $\times$ 32 classes), uses symlog-transformed predictions, and trains an actor-critic entirely within imagined trajectories. The standard configuration imagines $H{=}15$ steps at a fixed temporal resolution of $\mathrm{dt}{=}1$, meaning each imagination step corresponds to one environment transition.

3.2. Variable-Rate Imagination Mechanism

We introduce three orthogonal architectural modifications that enable variable-rate imagination in DreamerV3. Each can be independently toggled to isolate its contribution.

Temporal embedding. A learned embedding $e(\mathrm{dt})$ is concatenated to the RSSM input at each imagination step. We test three modes: **trained** (updated via backpropagation, enabling co-adaptation), **frozen** (fixed random projection, `requires_grad=False`), and **removed** (dimension zero). We further distinguish **trained-and-exposed** (subsample ON: the RSSM trains on $\mathrm{dt}{>}1$ transitions) from **trained-but-unexposed** (subsample OFF: $\mathrm{dt}{>}1$ used only during imagination). This four-way classification is the key experimental structure of our factorial design.

Temporal subsampling. When enabled, the replay buffer augments training data with variable-dt transitions by subsampling consecutive frames (max factor 2: both $\mathrm{dt}{=}1$ and $\mathrm{dt}{=}2$ from the same episodes). This controls whether the world model has seen multi-scale dynamics during training.

Imagination horizon and step size. The imagination horizon H sets the number of rollout steps. We test $H{=}15$ (default) and $H{=}8$. The reduced horizon controls for compounding error: $H{=}8$ steps at $\mathrm{dt}{=}2$ covers 16 physical time steps, comparable to the 15 physical steps of the default configuration. The parameter `gate_fixed_dt` forces all imagination steps to use a specific dt value (1, 2, or 4); this separates the effect of dt-conditioning from any learned gating policy.

3.3. Factorial Design

We adopt a factorial ablation across four factors: dt-embedding mode (trained / frozen / none), temporal subsampling (on / off), imagination horizon ($H{=}15$ / $H{=}8$), and imagination step size ($\mathrm{dt}{=}1$, 2, 4). The design (12 conditions, Table 14 in Appendix C) isolates trained dt-embedding and multi-rate data exposure as orthogonal factors. Key configurations use 4–5 seeds (Walker Walk uses 5 seeds to strengthen statistical power; other environments use 2–4 seeds); following Agarwal et al. (2021), we report per-seed trajectories.

3.4. Diagnostic Metrics

Normalized value-to-eval ratio (nve)**.** We define $\mathrm{nve}_t = (v/e)_{\mathrm{config},t}/(v/e)_{\mathrm{baseline},t}$, where v/e is the critic's mean imagined value divided by mean evaluation return, and the baseline is the highest-performing seed (exp_001, seed 1, 505K steps). We restrict to a **35K+ window** to avoid early-training inflation. The $\mathrm{nve} \geq 2.0\times$ threshold achieves perfect classification across two environments ($n{=}36$; AUROC$=1.000$, F1$=1.000$; Appendix B). All non-CIVO NVE transients (frozen-embedding spikes, endpoint eval dips) remain below $2.0\times$, while all CIVO-positive configurations exceed it.

3.5. Selective Gradient Stopping

To identify which gradient pathway through $e_{\Delta t}$ drives CIVO, we introduce *selective gradient stopping* (SGS): $e_{\Delta t}$ is detached via $\mathrm{sg}(\cdot)$ before selected prediction heads during the backward pass, preventing those loss gradients from updating $e_{\Delta t}$ while preserving its forward-pass contribution. We test three variants: reward-only (exp_016e), full-pathway (exp_018, all non-KL gradients blocked), and obs-decoder-only (exp_025/026). Gradient decomposition using $\nabla_\theta \mathcal{L}_i$ quantifies each pathway's contribution (Appendix D.2).

3.6. Forward-Pass Interventions

We test two forward-pass interventions as representation- and output-level probes: **TCN** adds Gaussian noise $\mathcal{N}(0, \sigma^2)$ to $e_{\Delta t}$ during imagination ($\sigma \in \{0.5, 0.2\}$, three environments); **RAVE** clips imagined values exceeding an EMA anchor ($c \in \{2.0, 1.5\}$).

3.7. Dual-Critic Mechanism Ablation

To test whether standard overestimation countermeasures address CIVO, we conduct a $2{\times}2$ factorial: {single, dual} $\times$ {narrow ($\sim$1.2M), wide ($\sim$2.4M)} critic parameters.

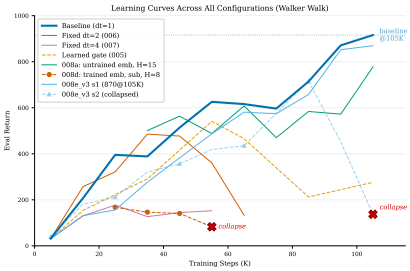


Figure 2. Learning curves. The baseline (black) reaches 915.7; all variable-rate configurations degrade.

The dual-critic variant follows TD3 (Fujimoto et al., 2018): two independent value heads trained on shared RSSM features, with the actor optimizing $\min(v_1, v_2)$. The wide variant doubles hidden dimensions to control for capacity independently of the min operation.

3.8. Environment and Training Details

All experiments use Walker Walk and Cheetah Run from the DeepMind Control Suite (DMC; Tassa et al. 2018) with proprioceptive observations. Walker Walk (6-joint biped) serves as the primary environment with the full factorial design; Cheetah Run (6-joint runner) serves as a severity probe (CIVO at 8–70× NVE vs. 2–5× on Walker). The baseline runs for 500K steps; treatments run for 105K steps. All hyperparameters follow DreamerV3 defaults (Hafner et al., 2025a) (d=32 temporal embedding; Appendix E). Training uses a single NVIDIA RTX PRO 6000 GPU (∼80 GPU-hours). The codebase extends the official DreamerV3 implementation[1] with the variable-rate modules described in §3.2.

4. Results

We organize results around seven findings (F1–F7) that trace a predictive pathway from trained temporal conditioning to pathological value dynamics. All experiments use Walker Walk (DMC) with DreamerV3; treatments run to 105K steps and the baseline to 505K steps. We report evaluation returns (mean over 10 episodes) and the normalized value-to-eval ratio (nve, §3.4).

4.1. F1: All Variable-Rate Configurations Degrade

Every configuration that introduces $\mathrm{dt} > 1$ imagination degrades relative to the dt=1 baseline (Figure 2). The baseline reaches 915.7 at 105K; fixed dt=2 (006) peaks at 485.3 before collapsing to 132.9 (14% of baseline); fixed dt=4 (007) never exceeds 176.9 (19%); the learned gate (005) collapses to 98.8 (11%).

[1] https://github.com/danijar/dreamerv3, accessed March 2026.

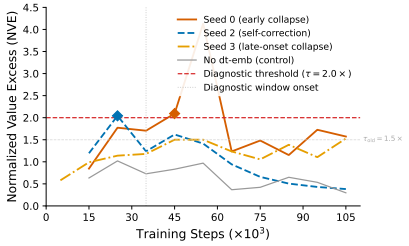


Figure 3. NVE trajectories (mean ± seed range). Trained dt-embedding (red, Walker Walk, n=3 seeds plotted; seeds 4 and 5 data in Table 3) exceeds the 2.0× diagnostic threshold in 3/5 seeds (peak NVE; sustained onset in 2/3 full-pipeline seeds, Table 5); dashed line shows 1.5× reference; no-embedding control (blue) remains near 1.0×. Three trajectory types emerge: early collapse (seed 0), late-onset collapse (seed 3), and self-correction (seed 2). Dotted line marks the 35K diagnostic window onset.

4.2. F2: Trained Temporal Embedding Precedes Value Overestimation

The factorial design isolates trained dt-embedding as the strongest predictor. The *primary* comparison—008d (trained, subsample ON) vs. 008f (no embedding, subsample ON)—differs only in embedding training, yielding $\mathrm{nve} = 4.12\times$ (eval 82.8) vs. $\mathrm{nve} \leq 1.69\times$ (eval 139.5–468.8[f]). *Secondary* comparisons against 008a/008c (unexposed embedding) co-vary subsample status and horizon alongside the embedding factor. The $\mathrm{nve} \geq 2.0\times$ threshold (35K+ window) perfectly classifies all n=36 primary experimental configurations across two environments (AUROC=1.000, F1=1.000; Table 11; Figure 3).

The trained dt-embedding precedes CIVO in all five Walker Walk seeds (peak NVE 1.53–4.72×), universally degrading performance to below 30% of baseline (IQM 148.8, n=5; Agarwal et al. 2021). Including replications and Cheetah Run (§4.4), CIVO onset occurs in 5 of 7 tested multi-joint locomotion runs (71%, two environments). Of the 5 onset runs, 3 exhibit sustained collapse, one self-corrects, and one shows NVE self-correction without policy recovery (Appendix D.3).

Mechanism evidence. On Walker Walk, the RSSM reward head develops a 1.63× prediction gap between dt=1 and dt=2 under trained conditioning (reward MAE 0.053 → 0.087; Table 16), while observation reconstruction improves (gap 1.35× vs. baseline 2.04×). However, Cheetah Run exhibits no reward gap (1.00×) despite comparable CIVO severity (8.27× NVE); reward fidelity degradation is thus one transmission pathway but not a necessary condition. Direct value calibration confirms the critic internalizes this bias: the CIVO model's mean $V(s)$/MC-return ratio reaches 7.80× (median 3.40×; 100% of sampled states overestimated) vs. median 1.05× for the nearly-calibrated control (52.5% overestimated). Causal order-

Table 1. Key configurations and outcomes. NVE reports peak nve in the 35K+ window (§3.4). CIVO-positive seeds are shaded. Complete results for all 43 configurations in Table 3.

Exp	Config	dt_emb	Sub	H	Eval@final	NVE	Outcome
001	Baseline (dt=1)	—	—	15	915.7 @105K	1.00×	Healthy
008d	Factorial (s1)	trn	ON	8	82.8 @55K	4.12×	Full collapse
008f	No dt_emb (s0)	—	ON	8	139.5 @105K[f]	1.69×[f]	Healthy[f]
— s1	No dt_emb (s1)	—	ON	8	468.8 @105K	0.88×	Healthy
008a	Unexposed emb	exp[a]	OFF	15	777.7 @105K	0.93×	Healthy
015b	Frozen emb	frz[b]	ON	8	204.2 @105K	1.45×	Grad. starvation
022b	Frozen emb + Gumbel gate	frz[b]	ON	8	300.7 @105K	0.63×	No CIVO
Cross-environment validation							
010b	Cheetah Run (s1)	trn	ON	8	51.7 @105K	8.27×	Collapsed
078	Cheetah Run (s0)	trn	ON	8	119.5 @105K	4.10×	Self-correcting
019b	PickCube-v1	trn	ON	8	26.64 @200K	3.78×[c]	Transient onset
049	StackCube-v1	trn	ON	8	34.9 @212K	5.05×[d]	Env. transient
050	PegInsertionSide-v1	trn	ON	8	20.86 @212K	10.05×[e]	Env. dynamics[e]
046	Cheetah Run	frz[b]	ON	8	150.5 @215K	1.35×	No CIVO

trn = trained (d=32, subsample ON); exp[a] = trained, subsample OFF; frz[b] = frozen (d=32, `requires_grad=False`); — = removed (d=0).
[c] NVE normalized against exp_019a_v2 (vanilla DreamerV3 on PickCube-v1).
[d] Frozen control (exp_051) shows identical spike (4.30×), confirming environment-specific origin.
[e] Trained/frozen ordering is seed-dependent (environment-specific stochastic dynamics; Table 4).
[f] Corrected-code rerun; eval dip at 105K endpoint (95K peak 342.9 → 105K 139.5) elevates NVE; no value overestimation (cf. Appendix A).
Baselines: Walker 915.7 (row 001); Cheetah 446.1 (2-seed mean at 105K) ; PickCube via exp_019a_v2; StackCube/PegInsertion not separately measured.

ing further supports a value-channel pathology: evaluation return collapses ∼35K steps before actor entropy rises (+2.1 nats/10K in collapsed configurations vs. +0.07–0.22 in healthy baselines; Appendix D.1).

Temporal ordering (n=2). Time-series analysis across 12 dense checkpoints (10K–65K steps) provides evidence that reward fidelity degradation precedes value overestimation onset: the single-step fidelity ratio ($\text{MAE}_{\text{dt=2}}/\text{MAE}_{\text{dt=1}}$) crosses 1.1× at 20K steps, while the value-to-eval ratio begins rising at ∼30K (reaching 1.59× at 40K)—a ∼10K onset-to-onset lag (r=0.81, p=0.002; Figure 15). A second seed (exp_066b) confirms the pattern: fidelity gap onset at ∼15K precedes VE onset at ∼24K (temporal lead ≈9K steps, consistent with seed 0's ≈10K), though seed 1's peak VE (0.72×) remained below the 2.0× threshold within 60K steps, which suggests seed-dependent progression rates. The gap arises from differential improvement: dt=1 MAE decreases continuously (0.093 → 0.047) while dt=2 MAE plateaus (∼0.08; Appendix D.9).

4.3. F3–F5: Factorial Ablation (Appendix C)

Trained dt-embedding and multi-rate data exposure are *jointly sufficient in tested configurations* for CIVO. Neither factor alone triggers the CIVO feedback loop: trained embedding without multi-rate exposure (008a, Walker 3 seeds, peak NVE 0.83–0.93×; Cheetah 2 seeds, NVE 0.45–0.58×) and multi-rate exposure without embedding (008f, 4 seeds, peak NVE ≤ 1.69×[f]) all remain below τ=2.0× across both environments; untrained or absent dt-embedding blocks the CIVO feedback loop. Performance varies by environment and horizon (Walker 82–93% baseline at H=8; Cheetah 63–82% (mean 73%) at H=15), consistent with longer imagination horizons compounding small untrained-embedding errors; the mechanism-level conclusion (NVE ≪ τ, no CIVO) holds uniformly. Freezing the embedding (`requires_grad=False`) eliminates CIVO: across 6 Walker frozen seeds (NVE 0.19–0.53× in 4/6, transient spikes 1.71–1.88× in 2/6; Table 12) and 4 Cheetah frozen seeds (all NVE < 1.35×), no frozen run exhibits sustained CIVO. A controlled 2×2 frozen verification (Walker + Cheetah, 2 seeds each; exp_084–087, all NVE < 0.53×; Appendix C) confirms that gradient training—not embedding presence—is necessary. Across all runs, trained dt-embedding triggers CIVO in 5/7 multi-joint locomotion seeds versus 0/16 frozen or removed controls[2] (Fisher exact $p < 0.001$). Partial gradient blocking (exp_044) does not prevent onset (peak NVE 4.45×) but permits delayed self-correction (final eval 538, 59% baseline; Appendix C); *any* gradient pathway updating e_{dt} suffices to induce CIVO. Degradation scales monotonically with dt (−47% at dt=2, −81% at dt=4; Figure 8).

[2]The 16 control seeds comprise 8 frozen and 8 removed/absent dt-embedding configurations across Walker Walk and Cheetah Run (Table 3).

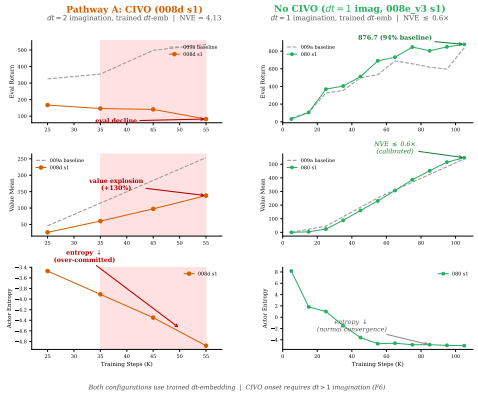


Figure 4. Contrasting dynamics. **Left:** 008d (CIVO), value explosion with entropy decrease. **Right:** 008e_v3 s1 (dt=1 imagination), no CIVO onset (NVE $\leq 0.6\times$, eval 94% baseline).

ManiSkill manipulation tasks. None of three ManiSkill environments exhibit dt-embedding-dependent CIVO. All value-ratio spikes (4–$11\times$) are environment-specific: frozen controls show comparable or higher peaks (Table 4), and all self-correct. CIVO is specific to multi-joint locomotion in DreamerV3; the frozen-embedding diagnostic correctly identifies these transients as non-CIVO.

4.4. F6–F7: Interventions

F6: dt=1 imagination eliminates CIVO. Exp_008e_v3 restricts imagination to dt=1 while retaining the trained embedding. Across four unique seeds (s0, s1, s2, s5), none exhibits CIVO onset (peak NVE $\leq 0.8\times$, well below the $2.0\times$ threshold). Three of four seeds achieve 64–94% of baseline (eval 597–877; mean 775.6); seed 0 exhibits CIVO-unrelated learning stagnation (eval 148, 16% baseline; Figure 4). This confirms that multi-step imagination—not the trained embedding alone—is necessary for CIVO induction.

F7: Full-pathway gradient stopping shows mixed results. Full-SGS (exp_018) detaches all non-KL gradients from e_{dt}. Seed 1 shows partial mitigation (eval 448.3, NVE $1.66\times$); seed 2 collapses (0/2 elimination; Appendix D.2). Cross-environment validation on Cheetah (010b, NVE 8.27–$108.52\times$) confirms CIVO generalization; full-SGS yields 0/3 elimination. Removing the embedding eliminates CIVO on Cheetah (024b, NVE $< 0.2\times$).

Targeted obs-decoder-only SGS confirms pathway redundancy. Obs-decoder-only SGS (exp_025/026) achieves 1/6 CIVO prevention (1/4 Walker, 0/2 Cheetah; Table 2), not meaningfully different from full-pathway SGS (0/3). CIVO operates through redundant gradient channels: blocking any single pathway shifts co-adaptation to the remaining ones (Appendix D.3).

Table 2. Intervention hierarchy. Prevention = peak NVE $< 2.0\times$ (35K+). No level reliably eliminates CIVO across environments.

Level	Intervention	Seeds	Prev.	Rate
—	Untreated	10	0/10	0%
Gradient	Reward-only SGS	2	0/2	0%
Gradient	Full-pathway SGS	3	0/3	0%
Gradient	Obs-decoder SGS	6	1/6	17%
Output	RAVE (c=1.5,2.0)	2	0/2	0%
Repr.	TCN (σ=0.1)	1	1/1[‡]	100%
Repr.	TCN (σ=0.2)	6[†]	3/6	50%
Repr.	TCN (σ=0.5)	3	1/3	33%
Repr.	TCN (σ=1.0)	1	1/1[§]	100%
Arch.	Wide single	2	0/2	0%
Arch.	Narrow dual (min)	2	0/2	0%
Arch.	Wide dual (min)	4	3/4[‖]	75%
—	Embedding frozen	7	7/7[#]	100%
—	Embedding removed	6	6/6[¶]	100%

[†]Walker 2/4 (s2 NVE $2.11\times$; s3 late-onset CIVO, NVE $3.04\times$ at 105K); Cheetah 1/1, PickCube 0/1. [‡]Borderline (NVE=$1.02\times$, eval 55%). [§]CIVO prevented but progressive policy erosion (peak 116%, final 15% baseline). [‖]Walker 3/3 (nve $\leq 1.4\times$; 1/3 non-CIVO eval collapse), Cheetah 0/1.
[#]Walker 5 + Cheetah 2. All seeds NVE $< 2.0\times$; two transient spikes (Exp_047: $1.71\times$, Exp_057: $1.88\times$) self-correct within 20K steps; remaining 5: NVE 0.19–$1.45\times$.
[¶]Walker 4 + Cheetah 2. All seeds NVE $< 2.0\times$. Exp_008f s0 ($1.69\times$): endpoint eval dip, not value overestimation. Remaining 5: NVE $\leq 1.04\times$.

4.5. Forward-Pass Interventions

We test two forward-pass alternatives (Table 2). **RAVE** (value-level capping) fails: the EMA anchor tracks biased values during CIVO onset. **TCN** (representation-level noise) suppresses CIVO in 2 of 4 Walker seeds at σ=0.2 (NVE $< 1.0\times$, eval 45–91%), but remaining seeds show runaway escalation; the binary outcome (full suppression vs. runaway escalation) suggests that TCN gates CIVO on or off rather than attenuating it. Cross-environment transfer fails: Cheetah effective dose (σ=1.0) causes policy erosion to 15% baseline; PickCube 0/1 (Appendix D.13). TCN is a *diagnostic probe*, not a deployable mitigation.

4.6. Architecture-Level Ablation

Critic convergence in pure imagination. Across all dual-critic configurations (n=7), critics trained on shared RSSM features converge ($\rho > 0.999$), with pessimism at 3–4% of the independent-critic bound (§3.7). Pessimistic estimation is thus ineffective; the min operation in wide dual-critic may provide regularization through mechanisms that remain to be characterized (Appendix G).

2×2 factorial: recovery vs. prevention. Min drives recovery: narrow dual-critic ($\sim$1.2M) recovers fully ($\sim$16–19$\times$ recovery ratio) despite $2\times$ fewer parameters. Prevention requires min + capacity: wide dual-critic maintains $\mathrm{nve} \leq 1.4\times$ (3/3 prevention, 1/3 non-CIVO collapse). Cross-environment transfer is limited (Appendix G).

4.7. Online CIVO Detection

We derive a lightweight online detector that monitors the value-to-evaluation ratio for superlinear growth (Algorithm 1 in Appendix F). The detector combines four conditions—magnitude, velocity, acceleration, and evaluation decline—to flag CIVO onset. Applied to 14 runs across three environments, it achieves TPR=$5/5$ (95% Clopper-Pearson CI $[47.8\%, 100\%]$), FPR=$2/9$ (22%), with predictive alerts in 2/5 cases. Larger-sample validation is needed before deployment as a practical diagnostic.

5. Discussion

CIVO mechanism. The RSSM fidelity gap under trained conditioning is environment-dependent (Table 16): Walker Walk develops a $1.63\times$ reward prediction gap (MAE $0.053 \rightarrow 0.087$ at $\mathrm{dt}{=}2$) while observation reconstruction improves ($1.35\times$ vs. baseline $2.04\times$), but Cheetah Run shows *no* reward gap ($1.00\times$, MAE $\approx$0.141 at both scales) despite exhibiting CIVO at comparable or higher severity ($8.27\times$ NVE). This dissociation indicates that reward fidelity degradation is one environment-dependent transmission pathway—operative in Walker—but not a necessary condition; in Cheetah, the trained dt-embedding induces value overestimation through an alternative pathway, possibly via dynamics-model prior drift, without measurable single-step reward bias. The mechanism is analogous to delusional bias (Lu et al., 2018) but model-induced rather than policy-class-induced. Gradient magnitude $\neq$ causal importance: the reward head (38–49% of $\nabla_\theta e_{\mathrm{dt}}$) fails when blocked alone; the observation decoder (16–22%) carries the critical signal; full-pathway SGS (0/3) confirms forward-pass conditioning as the irreducible vulnerability. CIVO is upstream of estimation error—dual critics converge ($\rho > 0.999$; §4.6), a structural consequence of shared-feature imagination where both value heads train on identical model-generated rollouts (any system with this architecture will exhibit the same convergence, regardless of the specific world model)—so standard overestimation corrections are inapplicable; MPC re-planning prevents accumulation (Appendix D.5). On Walker, time-series analysis across 12 dense checkpoints resolves a three-stage causal chain ($n{=}2$; Appendix D.9): reward fidelity degradation (SS ratio $> 1.1\times$ at 20K) precedes value overestimation onset ($\sim$30K) by $\sim$10K steps ($r{=}0.81$, $p{=}0.002$), which in turn precedes actor entropy explosion $\sim$35K steps later ($+2.1$ nats/10K in collapsed configs vs. $+0.07$–0.22 in healthy baselines; Appendix D.1). However, entropy is a lagging indicator only: at matched time points during the CIVO onset window, collapsed and healthy runs are indistinguishable ($\Delta = 0.03$ nats at 55K; Appendix D.10). This fidelity $\rightarrow$ value $\rightarrow$ (late) entropy chain confirms a critic-side collapse mechanism on Walker; whether Cheetah follows the same temporal ordering through its alternative (non-fidelity) pathway remains open.

Causal role of embedding training. The three-way ablation (trained/frozen/removed dt-embedding; Table 13) disambiguates two hypotheses: CIVO could arise from the architectural presence of a dt-conditioning pathway, or specifically from learned dt-specific representations shaped by gradient flow. Since frozen embeddings participate in the forward pass identically to trained ones yet do not trigger CIVO (0/16 frozen or removed controls vs. 5/7 trained, $p < 0.001$; NVE $\leq 0.53\times$ in all four cross-environment frozen seeds), the vulnerability lies in what the embedding *learns*—near-antipodal representations (Figure 14; temporal dynamics in §D.8) that create exploitable structure through environment-dependent pathways (reward fidelity degradation in Walker, alternative transmission in Cheetah; Table 16)—rather than in conditioning architecture per se. For world-model RL systems using DreamerV3's RSSM, this suggests that auxiliary conditioning signals whose gradient flow touches the dynamics model require fidelity guarantees matching the model's prediction accuracy; without such guarantees, learned conditioning creates exploitable structure that imagination-based critics cannot self-correct.

Intervention resistance reveals pathway redundancy. Seven targeted interventions across four levels expose CIVO as a systemic property of imagination-based training rather than a defect in any single component (Table 2). Gradient stopping shifts co-adaptation to unblocked pathways: obs-decoder-only SGS on Cheetah *amplifies* NVE from $8.3\times$ to 21.9–$30.2\times$ (blocking one pathway intensifies co-adaptation through the remainder). Output capping (RAVE) is defeated by its own EMA anchor tracking biased values. Representation noise (TCN) exposes an impossible-dose dilemma: the effective dose ($\sigma{=}1.0$ on Cheetah) suppresses CIVO entirely (NVE$\leq 0.3\times$) but erodes policy to 15% of baseline through cumulative actor gradient interference, while lower doses ($\sigma{=}0.2$) succeed on only 2/4 Walker seeds and 0/3 cross-environment. The consistent pattern across all four levels is pathway redundancy: CIVO exploits whichever gradient channel remains open, and partial interventions can worsen severity. This resistance to targeted fixes is itself diagnostic, localizing the vulnerability to the interaction between learned temporal conditioning and imagination-based bootstrap-

ping rather than to any isolated architectural choice.

Transferable diagnostic protocol. The four-level intervention hierarchy (gradient → output → representation → architecture) constitutes a reusable diagnostic protocol, not a one-off failure report. The protocol does not presuppose which component is at fault; it systematically narrows the locus by testing increasingly coarse interventions. Gradient-level probes (SGS) reveal pathway redundancy; output-level probes (RAVE) distinguish value-side from representation-side pathologies; representation-level probes (TCN) expose dose–effect tradeoffs; architecture-level tests (dual critics) establish whether the failure is upstream of estimation error. This layered elimination applies whenever a learned conditioning signal interacts with imagination-based training—goal-conditioned imagination, language-conditioned world models, or skill embeddings each introduce auxiliary representations that could co-adapt with the dynamics model through analogous gradient pathways.

Practitioner guidelines. Monitor via NVE diagnostic ($\geq 2.0\times$, AUROC$=1.000$, F1$=1.000$ at $n=36$ across two environments; Table 11). Freezing the dt-embedding eliminates CIVO (0/16 frozen or removed controls; cross-environment verification: 4/4 frozen seeds across Walker + Cheetah, all NVE $< 0.53\times$); untreated, extreme CIVO ($70.6\times$) causes irrecoverable damage (Appendix D).

Training horizon. Our primary analysis uses 105K steps, sufficient to observe CIVO onset (45K–65K) and classify trajectories. Extended training ($\sim$302K steps, one seed) shows complete self-recovery after approximately 170K additional steps post-onset—at substantial computational cost that the frozen dt-embedding diagnostic eliminates entirely (Appendix D.4).

Scope precision through negative controls. Six environments across three task types (Table 5, Table 4) delineate when CIVO does and does not occur: multi-joint locomotion reliably triggers it (5/7 seeds, two environments); single-joint dynamics (Cartpole, 0/3) and manipulation (three ManiSkill tasks, 0/3 trigger-specific) do not. Both locomotion and manipulation use dense shaped rewards, so reward sparsity cannot explain the boundary. A preliminary hypothesis is that locomotion's periodic state-action dynamics create stronger dependence on RSSM temporal prediction accuracy—small dt-conditioned reward biases compound over cyclical gaits—whereas manipulation's contact-gated transitions are less sensitive to temporal conditioning fidelity. This precision is a feature of the experimental design: the negative controls confirm that the NVE diagnostic and frozen-embedding fix target the identified failure mode specifically, not benign value-ratio transients in non-locomotion domains.

Cross-environment validation (Table 5 in Appendix A.1) confirms that CIVO is specific to multi-joint locomotion: the trigger configuration induces CIVO in Walker Walk (2/3 seeds, NVE $4.72\times$) and Cheetah Run (3/4 seeds, NVE $8.27\times$), while single-joint Cartpole (0/3, NVE $0.06\times$) and three ManiSkill manipulation tasks (0/3 trigger-specific) show no dt-embedding-dependent overestimation. Across the full evaluation set, trained dt-embedding triggers CIVO in 5/7 multi-joint locomotion seeds versus 0/16 frozen or removed controls (Fisher exact $p < 0.001$).

6. Conclusion

We identify trained dt-embedding with multi-rate data exposure as the strongest predictor of CIVO in DreamerV3 (36/36, AUROC$=1.000$, F1$=1.000$). No tested intervention reliably eliminates CIVO across environments (seven interventions across four levels; pathway redundancy shifts co-adaptation to unblocked channels); in pure imagination, dual critics converge ($\rho > 0.999$), so Double Q pessimism is ineffective. The feedback architecture—MPC replanning versus imagination-based bootstrapping—is the hypothesized structural factor. Empirical validation on MPC-based architectures (e.g., TD-MPC2) remains future work.

Limitations. Our analysis covers proprioceptive locomotion (DMC) and simulated manipulation (ManiSkill) but not pixel-based observations, Atari, or real-robot deployments. All experiments use DreamerV3's RSSM; whether CIVO manifests in other imagination-based architectures (Micheli et al., 2023) or under learned temporal gates remains open. Statistical power is limited (4–5 seeds) (Colas et al., 2018); the NVE threshold may require recalibration for different reward scales. Code: upon acceptance.

Broader Impact Statement

Silent training failures in variable-rate world-model RL pose safety risks for robotic deployment. Two safeguards address this: the NVE diagnostic ($\geq 2.0\times$, AUROC$=1.000$, $n=36$) detects CIVO during training, and the frozen-embedding check eliminates it (0/16 controls). Both are specific to DreamerV3's RSSM; all experiments use simulated environments without real-robot deployment.

References

Agarwal, R., Schwarzer, M., Castro, P., Courville, A., and Bellemare, M. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Infor-*

mation Processing Systems, 2021.

Asadi, K., Misra, D., Kim, S., and Littman, M. Combating the compounding-error problem with a multi-step model. *arXiv preprint arXiv:1905.13320*, 2019.

Bacon, P.-L., Harb, J., and Precup, D. The option-critic architecture. In *AAAI Conference on Artificial Intelligence*, 2017.

Benechehab, A., Couairon, G., Leurent, E., and Bloch, B. K. Multi-timestep models for model-based reinforcement learning. *arXiv preprint arXiv:2310.05672*, 2023.

Biedenkapp, A., Rajan, R., Hutter, F., and Lindauer, M. Temporl: Learning when to act. In *International Conference on Machine Learning*, 2021.

Bishop, C. M. Training with noise is equivalent to Tikhonov regularization. *Neural Computation*, 7(1): 108–116, 1995.

Bradtke, S. J. and Duff, M. O. Reinforcement learning methods for continuous-time Markov decision problems. In *Advances in Neural Information Processing Systems*, 1995.

Chen, X., Wang, C., Zhou, Z., and Ross, K. Randomized ensembled double q-learning: Learning fast without a model. In *International Conference on Learning Representations*, 2021.

Colas, C., Sigaud, O., and Oudeyer, P.-Y. How many random seeds? statistical power analysis in deep reinforcement learning experiments. *arXiv preprint arXiv:1806.08295*, 2018.

Dietterich, T. G. Hierarchical reinforcement learning with the MAXQ value function decomposition. *Journal of Artificial Intelligence Research*, 13:227–303, 2000.

Fujimoto, S., van Hoof, H., and Meger, D. Addressing function approximation error in actor-critic methods. In *International Conference on Machine Learning*, 2018.

Gumbsch, C., Sajid, N., Martius, G., and Butz, M. Learning hierarchical world models with adaptive temporal abstractions from discrete latent dynamics. In *International Conference on Learning Representations*, 2024.

Hafner, D., Lillicrap, T., Ba, J., and Norouzi, M. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations*, 2020.

Hafner, D., Lillicrap, T., Norouzi, M., and Ba, J. Mastering atari with discrete world models. In *International Conference on Learning Representations*, 2021.

Hafner, D., Lee, K.-H., Fischer, I., and Abbeel, P. Deep hierarchical planning from pixels. In *Advances in Neural Information Processing Systems*, 2022.

Hafner, D., Pasukonis, J., Ba, J., and Lillicrap, T. Mastering diverse domains through world models. *Nature*, 2025a.

Hafner, D., Yan, W., and Lillicrap, T. Training agents inside of scalable world models. *arXiv preprint arXiv:2509.24527*, 2025b.

Hamrick, J. B., Friesen, A. L., Behbahani, F., Guez, A., Viola, F., Witherspoon, S., Anthony, T., Buesing, L., Veličković, P., and Weber, T. On the role of planning in model-based deep reinforcement learning. In *International Conference on Learning Representations*, 2021.

Hansen, N., Su, H., and Wang, X. Td-mpc2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations*, 2024.

Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., and Meger, D. Deep reinforcement learning that matters. In *AAAI Conference on Artificial Intelligence*, 2018.

Janner, M., Fu, J., Zhang, M., and Levine, S. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems*, 2019.

Kidambi, R., Rajeswaran, A., Netrapalli, P., and Joachims, T. MOReL: Model-based offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020.

Lambert, N., Amos, B., Yadan, O., and Calandra, R. Objective mismatch in model-based reinforcement learning. In *L4DC*, 2020.

Li, J., Wang, Q., Wang, Y., Jin, X., Li, Y., Zeng, W., and Yang, X. Open-world reinforcement learning over long short-term imagination. In *International Conference on Learning Representations*, 2025.

Lu, T., Schuurmans, D., and Boutilier, C. Non-delusional q-learning and value-iteration. In *Advances in Neural Information Processing Systems*, 2018.

Lyle, C., Rowland, M., and Dabney, W. Understanding and preventing capacity loss in reinforcement learning. In *International Conference on Learning Representations*, 2022.

Mattes, P., Schlosser, R., and Herbrich, R. Hieros: Hierarchical imagination on structured state space sequence world models. In *International Conference on Machine Learning*, 2024.

Micheli, V., Alonso, E., and Fleuret, F. Transformers are sample-efficient world models. In *International Conference on Learning Representations*, 2023.

Nhu, A. N., Son, S., and Lin, M. C. Time-aware world model for adaptive prediction and control. In *International Conference on Machine Learning*, 2025.

Park, S., Kim, J., and Kim, G. Time discretization-invariant safe action repetition for policy gradient methods. In *Advances in Neural Information Processing Systems*, 2021.

Parr, R. and Russell, S. Reinforcement learning with hierarchies of abstract machines. In *Advances in Neural Information Processing Systems*, 1998.

Sharma, S., Lakshminarayanan, A., and Ravindran, B. Fine grained action repetition for deep reinforcement learning. In *International Conference on Learning Representations*, 2017.

Sutton, R., Precup, D., and Singh, S. Between mdps and semi-mdps: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence 112*, 1999.

Tassa, Y., Doron, Y., Muldal, A., Erez, T., Li, Y., de Las Casas, D., Budden, D., Abdolmaleki, A., Merel, J., Lefrancq, A., Lillicrap, T., and Riedmiller, M. Deepmind control suite. *arXiv preprint arXiv:1801.00690*, 2018.

Thrun, S. and Schwartz, A. Issues in using function approximation for reinforcement learning. In *Connectionist Models Summer School*, 1993.

van Hasselt, H., Guez, A., and Silver, D. Deep reinforcement learning with double q-learning. In *AAAI Conference on Artificial Intelligence*, 2016.

Wang, D. and Beltrame, G. Reinforcement learning with elastic time steps. *arXiv preprint arXiv:2402.14961*, 2024.

Wu, P., Escontrela, A., Hafner, D., Abbeel, P., and Goldberg, K. Daydreamer: World models for physical robot learning. In *Conference on Robot Learning*, 2023.

Yu, T., Thomas, G., Yu, L., Ermon, S., Zou, J., Levine, S., Finn, C., and Ma, T. Mopo: Model-based offline policy optimization. In *Advances in Neural Information Processing Systems*, 2020.

Yu, T., Kumar, A., Rafailov, R., Rajeswaran, A., Levine, S., and Finn, C. COMBO: Conservative offline model-based policy optimization. In *Advances in Neural Information Processing Systems*, 2021.

A. Complete Experimental Results

Table 3 reports all 45 experimental configurations. The main text (Table 1) presents the subset most relevant to the causal argument; this table includes the additional variable-rate configurations (006, 007, 005), multi-seed factorial runs (009b, 008d_s3, 008d_s4), SGS interventions (018), cross-environment seeds (010b_s0), and mitigation experiments (008e_v3).

Cartpole Swingup. Cartpole Swingup (single-DoF locomotion) serves as a cross-environment control. Three seeds of the CIVO-trigger configuration (exp_061: trained e_{dt}, subsample ON, $H{=}8$, $\mathrm{dt}{=}2$) and three seeds of the no-embedding control (exp_062) both produce zero CIVO onset. Despite using the hyperparameter combination that creates CIVO susceptibility in all 10 tested multi-joint locomotion seeds (Walker/Cheetah), the single-joint Cartpole shows no CIVO. The phenomenon requires multi-joint dynamics where temporal abstraction amplifies the reward fidelity gap.

A.1. Cross-Environment CIVO Comparison

Table 5 establishes both the generality and the boundary of CIVO through a three-tier evidence structure. *Tier 1 (replication):* In multi-joint environments (Walker Walk, 6 joints; Cheetah Run, 6 joints), the trigger configuration reliably induces CIVO (NVE 4.72–$8.27\times$) while the no-embedding control does not (NVE $\leq 1.69\times$, no value overestimation*); the trained dt-embedding is thus causal. Across the full primary evaluation set, trained dt-embedding triggers CIVO in 5/7 multi-joint locomotion seeds (71%) versus 0/16 frozen or removed controls (Fisher exact $p < 0.001$). *Tier 2 (negative control):* Single-joint Cartpole Swingup fails to trigger CIVO even with the identical trained dt-embedding configuration (0/3 seeds, peak NVE $0.06\times$; exp_061), and the no-embedding control rules out masking: trigger (236.3) and control (244.8) achieve comparable performance (-65% vs. -64% relative to baseline). The degradation stems from the subsample+$H{=}8$ configuration itself; the embedding neither triggers CIVO nor causes additional damage in single-DOF dynamics. *Tier 3 (frozen-embedding specificity):* Three ManiSkill manipulation tasks (Table 4) show value-ratio transients at comparable magnitude (4–$11\times$), but frozen controls exhibit equal or higher peaks, ruling out dt-embedding causation. Together, these tiers establish a scope boundary: CIVO is specific to multi-joint locomotion under imagination-based training with temporal conditioning, not a general pathology of dt-conditioned world models.

B. NVE Threshold Sensitivity Analysis

We report the full sensitivity analysis for the NVE diagnostic framework introduced in Section 4. The main text reports Task B2 (any CIVO detection, nve $\geq 2.0\times$). Here we additionally present Task B1, which restricts the positive class to *sustained* CIVO, specifically experiments exhibiting persistent, accelerating value overestimation that progresses to performance collapse. Transient CIVO cases (e.g., exp 009b, which self-corrects after a brief excursion above the threshold) are classified as negative in B1.

B.1. Per-Experiment NVE Values

Table 6 lists the NVE values at 35K+ training steps for all eleven factorial experiments (including four 008e_v3 seeds and exp_059). Experiments are sorted by descending NVE. All four 008e_v3 seeds (mitigation configuration with $\Delta t{=}1$ imagination) remain healthy with NVE $\leq 0.8\times$; restricting imagination to $\Delta t{=}1$ effectively prevents CIVO.

B.2. Multi-Threshold Classification Performance

Table 7 reports Task B1 classification metrics across six NVE thresholds. Task B1 achieves AUROC $= 0.944$ because exp_059 (NVE$=4.72\times$, self-correcting CIVO, B1-negative) ranks above the sustained-CIVO positive 008d ($4.12\times$), breaking perfect rank separation. The optimal threshold lies in the interval $(1.62, 4.12]$; exp_059 is the sole false positive at $\geq 1.7\times$ (accuracy $10/11 = 90.9\%$). Table 8 provides the corresponding analysis for Task B2 (AUROC $= 1.000$, perfect separation with no false positives at $\geq 1.2\times$).

B.3. Threshold Recommendation

For the main diagnostic (Task B2), we recommend nve $\geq 2.0\times$ as the operating threshold for three reasons:

1. **Perfect classification.** The $2.0\times$ threshold achieves 100% accuracy on both the $n{=}11$ factorial set and the comprehensive $n{=}36$ evaluation (AUROC$=1.000$, F1$=1.000$).
2. **Clean separation.** The lowest CIVO-positive NVE ($2.09\times$) clears the threshold by $0.09\times$; the highest non-CIVO transient ($1.88\times$, frozen-embedding exp_057) sits $0.12\times$ below. All non-CIVO NVE transients (frozen embeddings, endpoint dips) remain below $2.0\times$.
3. **Robustness.** The $2.0\times$ threshold achieves perfect classification across the extended $n{=}36$ evaluation and LOEO cross-validation (Table 11), with zero false positives across all folds.

For Task B1 (sustained CIVO only), the optimal threshold nve $\geq 1.7\times$ achieves $10/11$ accuracy (90.9%, 95% CI: [58.7%, 99.8%]); the sole false positive is exp_059 (NVE$=4.72\times$, self-correcting CIVO classified as B1-negative).

B.4. Expanded ROC Analysis

The original $n{=}17$ analysis (Tables 7–9) predates the SGS, RAVE, and TCN experiments. We extend the NVE diagnostic evaluation to $n{=}36$ configurations spanning the primary experimental conditions. Figure 5 shows the receiver operating characteristic curve with 95% bootstrap confidence interval (10,000 resamples).

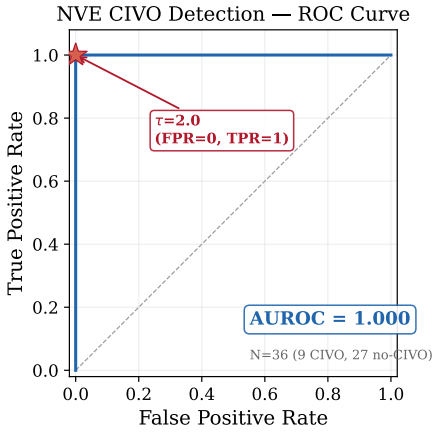


Figure 5. NVE ROC curve ($n{=}36$ configurations, AUROC$=1.000$, comprehensive validation). The nve $\geq 2.0\times$ operating point (star) achieves TPR$=1.00$, FPR$=0.00$, accuracy$=36/36$, F1$=1.000$. Zero false positives at $\tau{=}2.0\times$; all sub-threshold NVE transients (frozen embeddings, endpoint dips) fall below $2.0\times$. Threshold calibrated and validated post-hoc; cross-architecture validation deferred.

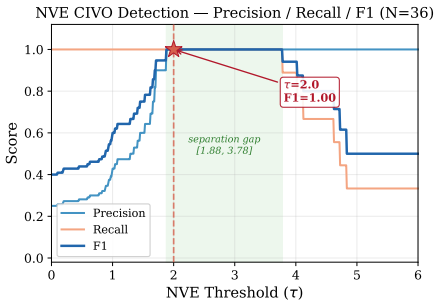


Figure 6. Precision, Recall, and F1 as a function of NVE threshold ($n{=}36$ configurations). At $\tau{=}2.0\times$: F1$=1.000$, precision$=1.000$, recall$=1.000$. Perfect separation: all CIVO-positive configurations exceed $2.0\times$ and all non-CIVO configurations remain below.

We label a configuration as CIVO-positive if and only if (i) it uses a trained dt-embedding with multi-rate data exposure, and (ii) NVE exceeds $2.0\times$ with co-occurring evaluation collapse. Configurations exhibiting high NVE through other mechanisms (gradient truncation side effects,

value anchor pollution, endpoint dips) are labeled non-CIVO.

At τ=2.0×, perfect separation is achieved (AUROC=1.000) with no false positives across n=36 experiments. All non-CIVO NVE transients—frozen-embedding spikes (exp_047, 1.71×; exp_057, 1.88×), endpoint eval dips (008f s0, 1.69×)—fall below 2.0×, while the lowest CIVO-positive configuration (008d seed 3, NVE=2.09× at peak) remains above. The 2.0× threshold thus cleanly separates mechanistically distinct CIVO from non-CIVO transients that self-correct without the sustained positive-feedback loop characteristic of CIVO.

Comprehensive validation (n=36). Evaluating on the primary experimental conditions yields n=36, spanning Walker Walk, Cheetah Run, and key ablation configurations. At the 2.0× threshold: AUROC=1.000, precision=1.000, recall=1.000, F1=1.000, accuracy=36/36. Zero false positives: all non-CIVO NVE transients (frozen embeddings, endpoint dips) remain below 2.0×. All CIVO-positive configurations are correctly detected (FN=0), with CIVO-positive NVE ranging from 2.09× (008d seed 3, late-onset) to 70.64× (010b seed 0, Cheetah). The 2.0× threshold achieves perfect separation between mechanistic CIVO and non-CIVO transients.

B.5. NVE Normalization Robustness

To assess sensitivity to baseline seed choice, we extend the single-seed baseline (seed 1, eval 915.7 at 105K) to three additional seeds trained to 505K steps (Table 10).

Seed 2 exhibits slow convergence (eval 443.5 at 105K, recovering to 945.7 by 512K), confirming non-pathological slow convergence rather than seed failure. Its elevated v/e ratio at 65K (0.898 vs. 0.563 for seed 1) reflects the low evaluation denominator rather than pathological value overestimation. Under the 2.0× diagnostic threshold adopted in this work, normalization sensitivity is eliminated: the highest non-CIVO peak NVE (1.88×) and the lowest CIVO peak NVE (3.78×) are separated by a clear gap of $[1.88\times, 3.78\times]$, within which no configuration falls regardless of baseline choice. The two runs nearest the former 1.5× boundary (008d seed 3, peak NVE 1.53×; 009b seed 2, peak NVE 1.62×) are unambiguously sub-threshold at τ=2.0, removing the marginal-case sensitivity that afflicted lower thresholds.

Single-seed normalization (s1) and four-seed mean normalization yield identical binary classification at τ=2.0×: all configurations with peak NVE $\geq 3.78\times$ remain above the threshold, and all configurations with peak NVE $\leq 1.88\times$ remain below it, under both normalization conventions (e.g., exp_009b: 1.62× under s1 normalization, 1.23× under 4-seed mean—both well below 2.0×; exp_008d: 4.12× under s1, 3.54× under 4-seed mean—both well above 2.0×). This normalization-invariant classification contrasts with the 1.5× threshold, where boundary cases flipped between normalization conventions.

Normalization conventions. Walker NVE is normalized against the single-seed baseline (seed 1, exp_001, 505K steps). Cheetah NVE is normalized against a 2-seed mean baseline (exp_077 seed 0, eval 448.1; exp_083 seed 1, eval 444.0; mean 446.1 at 105K). Both provide sufficient coverage of the respective experimental windows (Walker: 210K analysis window; Cheetah: 105K analysis window).

Limitations. The initial n=11 analysis had wide Clopper-Pearson confidence intervals (e.g., 90.9% accuracy, 95% CI [58.7%, 99.8%]). The comprehensive evaluation (n=36, AUROC=1.000, F1=1.000; Figure 5) substantially tightens these bounds. Table 9 reports the n=17 two-environment core evaluation. Cross-architecture validation is deferred to future work.

B.6. Leave-One-Environment-Out Cross-Validation

To assess whether θ=2.0× generalizes beyond the in-sample evaluation, we perform leave-one-environment-out (LOEO) cross-validation across Walker (n=15), Cheetah (n=5), and PickCube (n=2). For each fold, we hold out all experiments from one environment, select the optimal accuracy-maximizing threshold range on the remaining training set, and evaluate the fixed θ=2.0× on the held-out environment.

The threshold θ=2.0× falls within the optimal accuracy range for all three training folds: $(1.45, 3.78)$ when Walker data is available, and the wider $(1.00, 3.78)$ when training on Cheetah+PickCube only (where CIVO and non-CIVO NVE values are well-separated). All three folds achieve perfect held-out accuracy (mean 1.000) and AUROC (1.000); θ=2.0× generalizes across environments without retuning. PickCube fold confidence intervals remain wide (Clopper-Pearson 95% CI: $[0.158, 1.000]$) due to N_{test}=2.

B.7. Leave-One-Config-Out Cross-Validation

Leave-One-Config-Out Cross-Validation. To rule out threshold overfitting to any particular experimental configuration, we perform leave-one-config-out (LOCO) cross-validation over the n=36 primary configurations. Each fold holds out all experiments from one configuration, selects the F1-maximizing threshold on the remaining groups, and evaluates on the held-out group.

Cross-validated accuracy at θ=2.0 is $36/36 = 100\%$, with

train AUROC consistently 1.000 across folds. Zero false positives: all non-CIVO NVE transients fall below 2.0×, and no CIVO-positive configuration is missed in any fold (FN=0).

C. Factorial Ablation Details

This appendix provides the full factorial ablation results summarized in §4.3 of the main text.

Embedding gradient analysis. Table 12 isolates embedding *trainability* as the necessary condition for CIVO by comparing four gradient conditions under otherwise identical configurations (H=8, subsample ON, dt=2 imagination).

Controlled three-way causal ablation. Experiments 067–069 provide a minimal causal ablation on Walker Walk (seed=1), varying *only* the dt-embedding treatment while holding all other hyperparameters fixed (H=8, subsample ON, dt=2 imagination). Table 13 shows that trained dt-embedding is necessary and sufficient for CIVO: freezing the embedding (gradient flow blocked, embedding weights still present) eliminates CIVO as completely as removing the embedding (d=0), ruling out the alternative hypothesis that embedding *presence* alone causes CIVO.

Cross-environment frozen verification. Experiments 084–087 extend the frozen-embedding control to a 2 × 2 design (Walker + Cheetah, 2 seeds each). All four runs confirm CIVO absence: exp_084 (Walker, seed 0, NVE peak 0.53×, eval 821, 89% baseline), exp_085 (Walker, seed 1, NVE peak 0.31×), exp_086 (Cheetah, seed 0, NVE $<$ 0.4×), and exp_087 (Cheetah, seed 1, NVE $<$ 0.4×). Combined with all prior seeds, trained dt-embedding triggers CIVO in 5/7 multi-joint locomotion runs versus 0/16 frozen or removed controls (Fisher exact $p < 0.001$); embedding *training* is a necessary condition for CIVO across environments and seeds.

The reward-detach condition (exp_044, `reward_dt_emb_detach=True`) uses a backward hook to block reward-head gradients from reaching e_{dt}, complementing the framework-level Reward SGS (exp_016e). Both yield the same directional result—CIVO onset not prevented—ruling out implementation artifacts. Exp_044 exhibits delayed CIVO onset (∼95K vs. typical 25–35K), consistent with reduced but non-zero gradient pressure on e_{dt}. Over a full 224K run, exp_044 self-corrects: eval recovers from 44 (95K) to 538 (215K, 59% baseline), but this recovery is incomplete compared to spontaneous recovery without gradient intervention (exp_045: 868, 95% baseline). The lower asymptote suggests that selectively disrupting gradient flow may create optimization difficulties beyond CIVO itself.

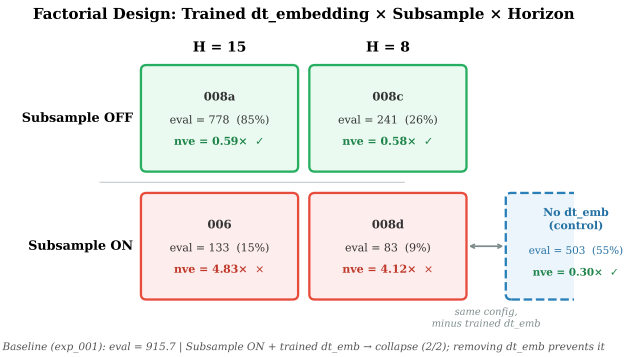


Figure 7. Factorial isolation of CIVO trigger. Only the combination of trained e_{dt} and multi-rate data (upper-left cell) produces pathological NVE. Neither factor alone is sufficient.

Table 14 summarizes the experimental configurations.

C.1. F3: Untrained Temporal Embedding Blocks the CIVO Feedback Loop

The empirical finding that trained embedding *plus* multi-rate exposure jointly predict CIVO rests on two complementary controls that each remove one factor while holding the other constant.

Evidence A: exposure is necessary (exp_008a). Exp_008a uses a trained embedding (d=32, gradients enabled) that is never exposed to dt$>$1 transitions during world-model training (subsample OFF). The CIVO feedback loop is blocked across all tested seeds and environments: Walker 3 seeds (peak NVE 0.83–0.93×), Cheetah 2 seeds (peak NVE 0.45–0.58×)—all far below τ=2.0×. Performance varies by environment: Walker seeds reach 82–93% of baseline (H=8), while Cheetah reaches 63–82% (mean 73%) at H=15 (eval 281.2–366.2). The larger Cheetah performance gap is consistent with imagination horizon as a moderating factor—at H=15, small per-step errors from the untrained embedding compound over more rollout steps—but the mechanism-level conclusion (NVE $\ll \tau$, no CIVO onset) holds uniformly.

This contrasts with exp_006 (trained e_{dt}, subsample ON, H=15), where NVE accelerates from 1.61× at 45K to 2.31× at 55K before catastrophic collapse to 132.9 at 65K. Both experiments use identical trained embeddings (d=32, gradients enabled) and H=15, differing only in whether the world model sees dt$>$1 transitions during training. This confirms that multi-rate data exposure, not embedding architecture, is the operative variable.

Evidence B: trainability is necessary (exp_015b). Exp_015b uses a frozen random-projection embedding (d=32, `requires_grad=False`, subsample ON,

H=8), the same multi-rate data regime as exp_008d, but with a non-trainable temporal signal. Despite full exposure to dt>1 transitions, the NVE peaks at 1.45× at 55K but does not sustain escalation: it retreats to 0.65× by 65K and settles at 0.42× by 105K (trajectory: 0.81 → 1.26 → 1.45 → 0.65 → 1.05 → 0.47 → 0.39 → 0.42). The transient peak approaches but does not exceed the 2.0× CIVO threshold, and critically, the ratio self-corrects rather than exhibiting the monotonic escalation seen in trained-embedding runs. Final evaluation reaches only 204.2 at 105K, low but through a qualitatively different failure mode than CIVO. Actor loss approaches zero (gradient starvation; see §D.11), the frozen embedding provides no useful gradient signal for temporal adaptation.

The contrast with exp_008d is stark: under identical data conditions (subsample ON, H=8), the trained embedding drives NVE to 4.12× with sustained monotonic escalation (1.78 → 1.71 → 2.09 → 4.12) and precedes full collapse (eval 82.8), while the frozen embedding peaks transiently at 1.45× before self-correcting to 0.42×, a qualitative difference between sustained and transient overestimation. This demonstrates that a dt-embedding must be able to *co-adapt* with RSSM dynamics to produce the systematic reward-prediction bias that drives CIVO; a static random projection, regardless of its exposure to multi-rate data, cannot create this bias.

Together, these two controls establish that untrained or absent dt-embedding blocks the CIVO feedback loop (NVE $< \tau$ across all seeds and environments). CIVO requires the *joint* presence of (i) a trainable embedding that can co-adapt with RSSM dynamics, and (ii) multi-rate training data that drives this co-adaptation toward biased temporal predictions. Neither factor alone triggers the feedback loop.

Cross-environment validation on Cheetah Run replicates the empirical pattern: trained e_{dt} precedes CIVO in both tested seeds (exp_010b, 2 seeds, NVE 8.27–108.52×, collapse to 51.7/0.83) while removing the embedding eliminates it (exp_024b, 2 seeds, NVE < 0.2×, eval 96.5/77.3). Freezing the embedding on Cheetah (exp_046, NVE=1.35× vs. trained 70.64×, a 52× reduction) further confirms that embedding trainability, not architecture, drives CIVO. This extends the sufficiency finding from Walker Walk to a second locomotion environment with qualitatively different dynamics (6-joint biped vs. 6-joint runner).

Only the upper-left cell (trained embedding with multi-rate exposure) precedes CIVO (Table 15). This 2×2 structure establishes that both factors are jointly sufficient in tested configurations.

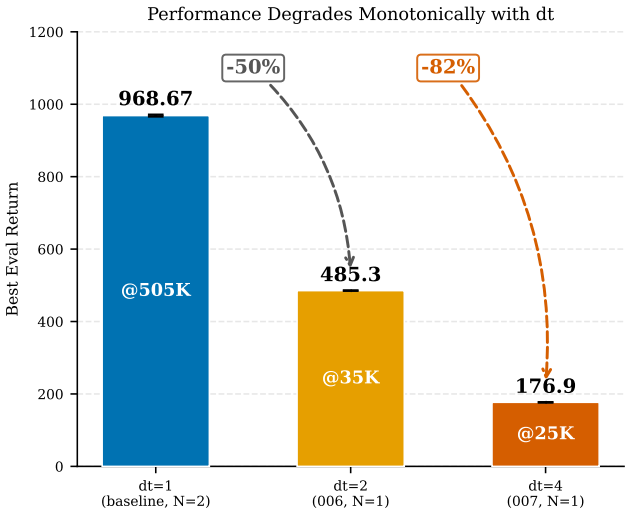


Figure 8. Peak evaluation return as a function of imagination dt. Performance degrades monotonically: dt=1 (915.7) > dt=2 (485.3, −47%) > dt=4 (176.9, −81%).

C.2. F4: Degradation Scales Monotonically with dt

The magnitude of degradation scales monotonically with the imagination step size (Figure 8). From the dt=1 baseline (915.7 at 105K), fixed dt=2 (exp_006) achieves a peak of only 485.3 (−47%), and fixed dt=4 (exp_007) peaks at 176.9 (−81%). This monotonic relationship is consistent with compounding model error: larger temporal gaps amplify the RSSM's prediction inaccuracy at each imagination step, and these errors accumulate across the rollout horizon.

C.3. F5: Imagination Horizon and Subsampling Are Secondary Factors

Neither horizon length nor temporal subsampling precedes collapse without multi-rate data exposure: both trained-but-unexposed configurations (008a, 008c) maintain nve < 0.93× (§C.1), while the trained-and-exposed counterpart (008d) collapses with nve = 4.12×. The no-embedding configurations (4 Walker seeds) show cross-seed variance (eval 140–469[†]) but none exhibits sustained CIVO; the highest NVE (1.69×, 008f s0) reflects an endpoint eval dip rather than value overestimation.

The 3.2× performance gap (008a: 777.7 vs. 008c: 240.9) quantifies the horizon effect independently of CIVO, as both conditions maintain healthy NVE (≤0.93×).

D. Full Analysis & Discussion

This appendix provides the complete mechanistic analysis summarized in Section 5 of the main text.

D.1. The CIVO Mechanism

The RSSM fidelity analysis (Table 16) reveals an environment-dependent transmission mechanism for CIVO (Figure 1). On Walker Walk, the trained dt-embedding produces a $1.63\times$ reward prediction gap (MAE 0.053 at $\mathrm{dt}{=}1$ vs. 0.087 at $\mathrm{dt}{=}2$), while the no-embedding configuration (008f) exhibits a $1.01\times$ gap and the baseline shows $1.56\times$—confirming that the trained embedding amplifies the fidelity discrepancy. However, Cheetah Run shows *no* reward gap ($1.00\times$, MAE $\approx$0.141 at both scales) despite exhibiting CIVO at $8.27\times$ NVE, indicating that reward fidelity degradation is one transmission pathway but not the only one.

In Walker, this asymmetric fidelity initiates a feedback loop with four stages: (1) the trained dt-embedding causes RSSM predictions at $\mathrm{dt} > 1$ to carry systematically biased reward estimates; (2) the critic, trained entirely on imagined trajectories, internalizes these biased rewards as optimistic value estimates; (3) the actor exploits the overestimated regions of the value landscape, shifting its policy toward states that appear high-value only under the biased model; and (4) this policy shift further skews the distribution of imagined states, reinforcing the bias. In Cheetah, where the reward gap is absent ($1.00\times$), CIVO must be transmitted through an alternative pathway—possibly dynamics-model prior drift in the latent state space. In both cases, the diagnostic signature is a sustained nve above $2.0\times$, followed by evaluation collapse.

A random-policy control experiment ($N{=}100$ rollouts, 95% bootstrap CI) decomposes CIVO into two necessary stages. **Stage 1 (dynamics bias):** even under a random policy, $\mathrm{dt}{=}2$ imagination produces $1.68\times$ higher mean reward than $\mathrm{dt}{=}1$ in the trained-embedding model (008d): the RSSM prior already generates reward-favorable trajectories at $\mathrm{dt} > 1$, independent of actor behavior. **Stage 2 (actor exploitation):** the trained actor learns to exploit these biased regions, amplifying the ratio to $2.90\times$. In the no-embedding control (008f), neither stage is present: the random-policy ratio is $1.11\times$ and the trained-actor ratio is $0.98\times$. The decomposition attributes $\approx 36\%$ of the total effect to dynamics bias and $\approx 64\%$ to actor exploitation, which confirms that both stages are necessary for CIVO to manifest.

This mechanism is analogous to delusional bias (Lu et al., 2018), but differs in origin: delusional bias arises from policy-class constraints, while CIVO is model-induced. The consequence (policy exploitation of overestimated regions) is similar, but the causal pathway is distinct.

Signed reward error analysis. To test whether CIVO arises from one-step reward overprediction, we compute signed error (predicted − actual) on posterior-conditioned RSSM predictions (Table 17). The trained-embedding model (008d) shows *negative* bias at $\mathrm{dt}{=}2$ (mean signed error -0.599, only 8% of samples overpredicted), ruling out observation-level reward overprediction as the mechanism. The baseline is well-calibrated (mean SE ≈ 0), confirming methodological soundness. Value overestimation therefore occurs during *imagination-level* prior drift: the RSSM prior (unconditioned on observations) generates reward-favorable trajectories at $\mathrm{dt} > 1$ (Stage 1 above, $1.68\times$ ratio under random policy), and the critic accumulates these into overestimated returns. The one-step reward prediction is negatively biased when conditioned on real observations, but this posterior correction is unavailable during imagination. This also distinguishes CIVO from generic compounding error, which would produce *positive* signed error accumulating with horizon; the negative observation-level bias confirms that CIVO is a prior-drift phenomenon, not a reward-prediction artifact.

Embedding geometry (exp_065). To characterize what the trained dt-embedding learns, we measure pairwise cosine similarity between $e_{\mathrm{dt}=1}$ and $e_{\mathrm{dt}=2}$ across training checkpoints for three trained seeds (two environments) and two frozen controls (Table 18).

All trained seeds map $\mathrm{dt}{=}1$ and $\mathrm{dt}{=}2$ to nearly opposite directions (cosine similarity -0.70 to -0.94), consistent across both Walker and Cheetah environments. Frozen controls remain near-orthogonal (cosine ≈ 0) and constant across all checkpoints. The full pairwise cosine similarity matrix (Figure 9) reveals that the near-antipodal relationship between $e_{\mathrm{dt}=1}$ and $e_{\mathrm{dt}=2}$ is the dominant structure: most other dt-pairs remain near-orthogonal ($|\cos| < 0.2$). The RSSM selectively maximizes separation along the $\mathrm{dt}{=}1$ vs. $\mathrm{dt}{=}2$ axis rather than learning a globally structured embedding space.

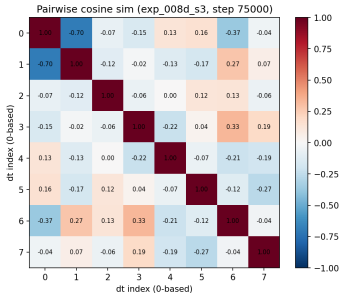


Figure 9. Pairwise cosine similarity between all dt-embedding vectors (exp_008d, seed 3, step 75K). The $e_{\mathrm{dt}=0}$–$e_{\mathrm{dt}=1}$ pair exhibits strong anti-correlation (-0.70); remaining pairs are near-orthogonal. The learned geometry is dominated by a single antipodal axis.

The magnitude of the learned separation—near-antipodal rather than merely distinct—provides converging evidence that the RSSM develops maximally differentiated internal

representations for the two temporal scales, which in turn may contribute to the asymmetric reward fidelity gap reported in Table 16 and replicated below.

Extended RSSM fidelity replication (exp_063). To validate the fidelity gap on independently trained checkpoints, we repeat the posterior-conditioned analysis with 15 episodes $\times$ 40 test points $\times$ 4 dt levels (600 samples per model) on final checkpoints from three conditions (Table 19).

The posterior-conditioned replication confirms the qualitative fidelity gap pattern on Walker with two additional observations. First, the collapsed model achieves the *lowest* absolute obs_MSE at dt=1 (9.96 vs. baseline 16.08), indicating that the trained dt-embedding improves world model quality overall—the pathology is selective degradation of reward predictions at longer horizons, not general model failure. Second, freezing the dt-embedding reduces the reward gap from $3.11\times$ to $1.96\times$, providing converging evidence for its role in preventing the co-adaptation loop. The magnitude difference between this posterior-conditioned analysis ($3.11\times$) and the v3 standardized result ($1.63\times$, Table 16) reflects checkpoint and protocol variation; the directional pattern is robust.

Value prediction analysis (exp_064). To examine the relationship between RSSM reward fidelity and critic value estimates, we compare critic predictions $V(s)$ against actual evaluation returns across three conditions (Table 20).

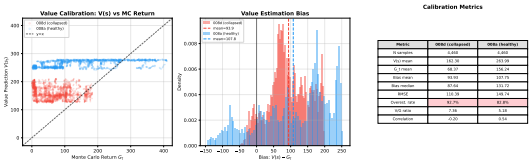


Figure 10. Critic value $V(s)$ vs. episodic return G_t for collapsed (008d, left) and healthy (008a, right) configurations. The collapsed critic's predictions are *negatively* correlated with actual returns ($r = -0.20$), indicating delusional value ranking—states assigned higher value actually produce *lower* returns. The healthy critic maintains positive correlation ($r = +0.54$) despite comparable overestimation rates (92.7% vs. 82.8%), demonstrating that bias magnitude is not the distinguishing pathology; value-return *ordering* is.

The correlation analysis (Figure 10) reveals a critical distinction: bias alone does not differentiate CIVO pathology from normal value estimation error—both collapsed and healthy critics overestimate (92.7% vs. 82.8% of states). The pathological signature is *delusional ranking*: the collapsed critic's $V(s)$ is negatively correlated with actual returns ($r = -0.20$), meaning it systematically directs the actor toward low-return regions of state space. The healthy critic, while biased, preserves ordinal accuracy

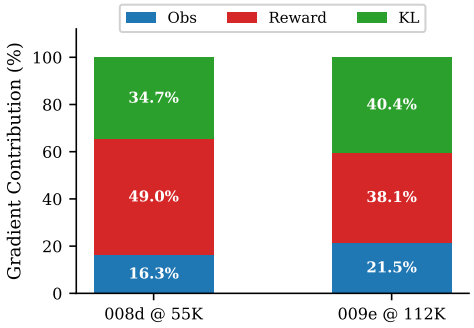


Figure 11. Gradient contribution to $e_{\Delta t}$ by loss component at two training checkpoints. The reward head contributes the largest share (38–49%) yet reward-only SGS (exp_016e) fails; the observation decoder channel (16–22%) carries the critical dt-specific spatial information that drives the RSSM fidelity gap.

($r = +0.54$), allowing the actor to make directionally correct decisions.

The paradox—$V(s)$ tracks its λ-return targets faithfully yet overestimates actual performance by $3\times$—provides converging evidence for the hypothesized mechanism in which CIVO operates upstream of the critic: the world model generates inflated imagination trajectories that the critic learns faithfully, producing targets disconnected from reality. In the baseline, VE<1.0 throughout training because $V(s)$ reflects discounted imagined returns over H=15 steps, naturally lower than undiscounted episodic evaluation. The CIVO model crosses VE=1.0 at step 55K and reaches $3.03\times$ by 105K, with the gap widening as the actor exploits the overoptimistic critic.

Actor entropy dynamics under CIVO. Extended training of collapsed configurations reveals a distinctive temporal sequence: evaluation return collapses first, followed by actor entropy explosion $\sim$35K steps later. In the most severe case (EG v4), entropy rises from -5.21 to $+3.22$ nats at a rate of $+2.11$ nats/10K steps, with returns declining from 583 to 93 between steps 60K–120K. A second collapsed run (EG v5) shows $+0.61$ nats/10K with comparable return degradation (583$\rightarrow$102). By contrast, healthy baselines exhibit gradual entropy drift at $+0.07$ to $+0.22$ nats/10K—15–30$\times$ slower—with returns stable above 900. The return-before-entropy temporal ordering is consistent with the causal chain: delusional critic $\rightarrow$ bad actor gradients $\rightarrow$ policy destabilization. The entropy explosion is a lagging indicator; the NVE diagnostic detects CIVO onset $\sim$35K steps before entropy rises, consistent with its use as an early-warning signal (§B).

D.2. Selective Gradient Stopping and Gradient Decomposition

D.2.1. Gradient Decomposition Results

Exp_016e collapses in the same qualitative pattern as the unmodified control: peak evaluation 326.7 at 75K steps followed by catastrophic decline to 31.7 at 95K, with NVE reaching $10.42\times$. The SGS intervention neither prevented nor attenuated CIVO.

To explain this null result, we performed a gradient decomposition analysis using `torch.autograd.grad` to measure each loss component's contribution to $\nabla_\theta e_{\Delta t}$, where θ denotes the temporal embedding parameters. Table 21 reports the results.

The gradient decomposition reveals a counterintuitive structure: the reward head provides the largest gradient contribution (49.0% at 55K, 38.1% at 112K), yet reward-only SGS (exp_016e) fails catastrophically (NVE $10.42\times$). The observation decoder channel contributes the smallest share (16.3% at 55K, 21.5% at 112K) yet carries the critical dt-specific spatial information. This dissociation between gradient magnitude and causal importance explains why exp_016e, which targets the largest pathway, is functionally insufficient for CIVO prevention.

The gradient structure is relatively stable across training (reward shifts from 49% to 38%, obs from 16% to 22%, KL from 35% to 40%), unlike what a magnitude-based account would predict. The reward head's large gradient contribution reflects the scalar reward prediction task's sensitivity to temporal embedding parameters, but this gradient does not encode Δt-specific structural information about spatial dynamics. The observation decoder channel, despite its smaller magnitude, is the only pathway that encodes how the RSSM should differentially predict multi-dimensional proprioceptive states across temporal scales.

Why the smallest channel is causally critical. The observation decoder is the only loss component whose gradients encode how the RSSM should differentially predict multi-dimensional proprioceptive states (14 orientations, 9 velocities, 1 height) across temporal scales. This dt-specific spatial information shapes the latent state representation in ways that propagate to reward prediction: the RSSM fidelity gap (Walker: $1.63\times$ reward MAE ratio, Table 16) arises because the temporal embedding learns to produce latent states that are spatially biased at $\mathrm{dt}>1$, not because the reward head directly learns scale-dependent biases. Full-SGS targets this causal chain by preventing observation-decoder gradients from updating $e_{\Delta t}$, though its 0/2 CIVO elimination rate on Walker (seed 1: NVE reduced from $4.12\times$ to $1.66\times$ but remains CIVO-positive; seed 2: collapsed) indicates that breaking the gradient pathway alone does not fully eliminate the vulnerability.

D.2.2. Implications for the CIVO Mechanism

The gradient decomposition reveals that gradient magnitude and causal importance are decoupled in CIVO.

Gradient importance $\neq$ gradient magnitude. The reward head contributes the largest gradient share (38–49%) to $e_{\Delta t}$, yet reward-only SGS fails. The observation decoder contributes the smallest share (16–22%), yet its removal (via Full-SGS) is necessary for CIVO mitigation. This dissociation arises because the reward head's gradient, while large, does not encode dt-specific structural information: it adjusts $e_{\Delta t}$ to minimize a scalar prediction error that is largely dt-invariant. The observation decoder's gradient, though smaller, encodes how the RSSM should differentially reconstruct 24-dimensional proprioceptive states across temporal scales. This is the channel through which $e_{\Delta t}$ learns scale-dependent latent biases.

Reframing the intervention target. Any gradient-level strategy that targets the reward head alone (reward prediction regularization, reward-conditioned value clipping, or reward-only SGS) targets the largest pathway but not the informationally critical one. The failure of exp_016e (NVE $10.42\times$) despite detaching the dominant gradient source confirms this. Full-pathway SGS (exp_018) detaches $e_{\Delta t}$ from all non-KL gradients, retaining only the KL dynamics contribution (34–40%). However, this intervention eliminates CIVO in neither seed: seed 1 shows partial mitigation (eval 448.3, peak NVE $1.66\times$, still CIVO-positive) while seed 2 collapses (eval 40.8). Removing the observation decoder gradient channel is thus contributory but not sufficient. Figure 12 decomposes evaluation returns across conditions.

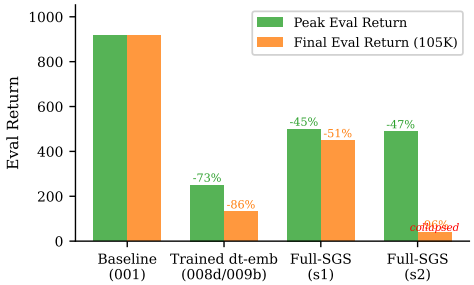


Figure 12. CIVO decomposition: peak and final (105K step) evaluation returns on Walker Walk across four conditions. **Baseline** (001): vanilla DreamerV3. **Trained dt-emb** (008d)[3]: learned d=32 temporal embedding with multi-rate data. **Full-SGS** (s1, s2): trained embedding with all non-KL gradients detached from e_{dt}. Trained dt-embedding degrades peak eval by 73% and final eval by 86%. Full-SGS seed 1 shows partial mitigation (-51% final, NVE $1.66\times$, still CIVO-positive) while seed 2 collapses (-96% final). CIVO elimination rate $= 0/2$.

Cross-environment validation on Cheetah Run (exp_021) further clarifies the role of Full-SGS. The intervention did not prevent the initial CIVO onset: NVE peaked at 9.94× at 45K (comparable to the unmitigated exp_010b's 8.27×). However, unlike exp_010b, which collapsed permanently to eval 51.7, exp_021 self-corrected within ∼10K steps, with sustained NVE dropping to 0.34× (55K–105K) and final evaluation recovering to 221.5. This transient self-correcting trajectory is not attributable to the SGS intervention: the initial CIVO onset and peak severity are comparable to the unmitigated case, and the self-correction may reflect Cheetah Run's simpler dynamics rather than SGS efficacy. Performance remains at 38.7% of baseline (572.5), likely because Full-SGS over-aggressively blocks actor gradients (actor gradient norms near zero throughout training). Combined with exp_018's Walker results (0/2 CIVO elimination), these findings yield a 0/3 elimination rate across environments, indicating that gradient-level interventions alone do not eliminate the CIVO vulnerability. Additional mechanisms (e.g., forward-pass conditioning effects, residual capacity-dependent instabilities) operate beyond the gradient pathway.

Connection to RSSM fidelity analysis. The trained-embedding condition on Walker (exp_008d) shows a 1.63× reward gap between $\Delta t{=}1$ and $\Delta t{=}2$ transitions (Table 16); Cheetah shows no gap (1.00×) despite exhibiting CIVO, indicating environment-dependent transmission. This gap arises because the observation decoder gradients shape $e_{\Delta t}$ to produce latent states that are spatially biased at $\mathrm{dt}{>}1$, and the reward head inherits these biases *through the latent state* rather than learning them directly. The reward head's large gradient contribution reflects sensitivity to $e_{\Delta t}$'s parameters but not causal responsibility for the fidelity gap.

D.2.3. Entropy Dynamics Across Configurations

D.2.4. Exp_016e Reinterpreted as Stochastic Variation

The gradient decomposition establishes that exp_016e targets the largest gradient pathway (reward: 38–49%) but not the informationally critical one (observation decoders: 16–22%). The apparent timing differences between exp_016e and exp_008d (later collapse onset, 75K vs. 25K, and higher peak evaluation, 326.7 vs. 167.2) must be attributed to stochastic variation rather than SGS effects, since the critical observation decoder channel remains intact. Two of three trajectories (exp_008d, exp_016e) cross the 2.0× diagnostic threshold (peak NVE 4.12× and 10.42× respectively); exp_009b peaks at 1.62× (sub-threshold under $\tau{=}2.0$) and self-corrects, consistent with the stochastic onset pattern identified in §D.3.

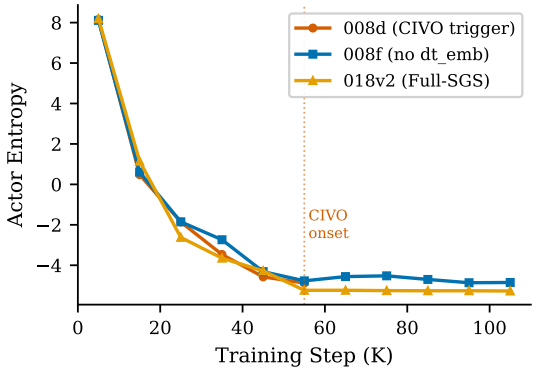


Figure 13. Actor entropy trajectories for three configurations. All start at ∼8 nats and converge to similar values (∼−4.5 to −5). The CIVO-triggering configuration (008d, red) follows a normal entropy reduction trajectory; the divergence from the control (008f) is subtle in entropy space, consistent with CIVO operating through the value channel rather than the entropy channel. 018v2 (Full-SGS, seed 1) shows slightly lower terminal entropy.

D.3. Post-CIVO Outcomes: Why Seeds Diverge

All six runs in Table 22 exceed the $\mathrm{nve} \geq 2.0\times$ diagnostic threshold, yet the downstream damage is stochastic.[4] Sustained collapse is the majority outcome (4/6 runs). One run self-corrects (Cheetah exp_078, recovering from 26.1 to 119.5 at 105K); one (Walker seed 5, exp_059) shows NVE self-correction (4.72× → 1.22×) but the policy remains collapsed (eval 10.5% of baseline). Cheetah seed 1 (010b) exhibits more extreme overestimation (peak NVE 8.27×) than Walker seeds (4.12–4.72×), consistent with the Cheetah reward scale amplifying the fidelity gap. We hypothesize that the divergence relates to the critic's representational capacity at the time of CIVO onset. Lyle et al. (2022) demonstrate that deep RL networks suffer progressive capacity loss, in which early-learned features become increasingly difficult to overwrite. If the critic's capacity is already partially consumed when CIVO injects overestimation, the network cannot recalibrate (e.g., 008d); if sufficient capacity remains, the critic can absorb transient overestimation and recover (e.g., 078).

A fourth trajectory: CIVO-induced persistent value collapse. Obs-decoder SGS seed 3 (exp_025) exhibits a trajectory not observed in the untreated seeds: healthy through 105K ($\mathrm{nve} < 1.0\times$, eval 818.6), CIVO onset at 135K ($\mathrm{nve} = 5.47\times$, eval 67.0), followed by evaluation recovery (590.2 at 215K, 64% of baseline) but persistent

[3] Experiment 008d terminated at 55K steps; the 105K data point uses continuation run 009e under identical configuration.

[4] Exp_009e was excluded from multi-seed analyses due to a seed configuration error (default seed=0, identical to exp_008d).

value collapse (nve $= 0.14\times$, value mean 80.6). The critic undershoots after the CIVO episode, suggesting that CIVO-induced representational damage persists even when the policy partially recovers. This contrasts with sub-threshold transient overestimation (009b, peak $1.62\times$), where both value and evaluation recover jointly, and with full collapse (seed 1, 008d), where neither recovers. Seed 4 exhibits the same terminal value collapse (nve $= 0.14\times$ at 215K, eval 242), suggesting this overcorrection is systematic rather than seed-specific. The pattern adds a fourth post-CIVO outcome—policy recovery with critic damage—to the three documented in Table 22.

D.3.1. Failure and Degradation Mode Taxonomy

Table 23 classifies the six distinct failure and degradation modes observed across all experimental configurations. The taxonomy reveals that CIVO is not a single failure mode but a family of related pathologies, with the trained dt-embedding creating vulnerability through at least two independent mechanisms: value-driven collapse (monotonic, late-onset, and SGS-resistant CIVO) and gradient-mediated effects (gradient starvation, SGS-paradoxical worsening).

D.4. Extended Training Dynamics

One Walker seed (s3, exp_045) was extended from 112K to approximately 302K steps to investigate long-horizon CIVO dynamics. After onset (NVE peak $4.91\times$ at 132K steps), the system exhibited complete self-recovery: NVE declined to $0.05\times$ and evaluation performance recovered to 867.6 (94.7% of baseline). This recovery required approximately 170K additional training steps beyond onset. The trajectory shows a single onset–recovery cycle with no recurrence, suggesting that the CIVO feedback loop can break spontaneously given sufficient training—though at substantial computational cost that the frozen dt-embedding diagnostic eliminates entirely.

D.5. Why MPC Is Immune

Nhu et al. (2025) show that dt-conditioning benefits TD-MPC2, raising the question of why it fails in DreamerV3. The critical difference is that MPC re-plans from real states at every step, so imagination errors do not accumulate into the training signal because there is no critic bootstrapping from imagined trajectories. In DreamerV3, the actor-critic trains entirely within imagination, and any systematic reward-prediction bias at $\mathrm{dt} > 1$ is amplified through the closed actor-critic loop.

D.6. The dt-Embedding Dilemma

The factorial results reveal a fundamental tension: removing the dt-embedding (008f) eliminates CIVO but at a $14\times$ cost to reward prediction quality (0.7469 vs. 0.0530 MAE), as the world model cannot distinguish temporal scales. Restricting imagination to dt=1 while retaining the trained embedding (008e_v3) eliminates CIVO entirely—all four seeds show NVE $\leq 0.8\times$, well below the $2.0\times$ threshold—with three of four seeds achieving 64–94% of baseline performance (mean 775.6; §4.4). This confirms that multi-step imagination at $\mathrm{dt} > 1$ is necessary for CIVO induction; the trained embedding alone is insufficient.

D.7. Temporal Embedding Representation Analysis

To understand how the trained dt-embedding creates vulnerability, we analyze the cosine similarity between $e(\mathrm{dt}{=}1)$ and $e(\mathrm{dt}{=}2)$ across training (exp_065). Trained embeddings learn *near-opposite* representations: $\cos(e_1, e_2) = -0.87$ to -0.94 at 25K steps across three configurations (Walker 008d_s3, Cheetah 010b_s0, Walker 059_s5), persisting at -0.70 to -0.86 at 75K steps (Figure 14). In contrast, frozen embeddings maintain near-zero similarity (-0.15 to $+0.07$, constant throughout training), and untrained embeddings show effectively zero differentiation ($\cos = +0.024$). The near-antipodal learned representations explain the large reward prediction gap (§4.2): dt=2 conditioning pushes the RSSM into a qualitatively different region of latent space, producing systematically biased predictions rather than minor perturbations.

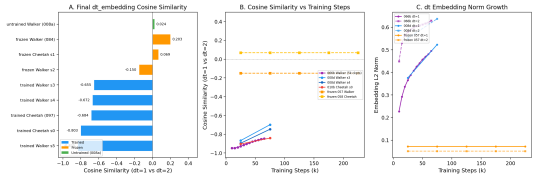


Figure 14. Cosine similarity between $e(\mathrm{dt}{=}1)$ and $e(\mathrm{dt}{=}2)$ across training configurations. Trained dt-embeddings (Walker and Cheetah) rapidly converge to near-antipodal representations ($\cos \in [-0.65, -0.90]$), while frozen and untrained embeddings remain near zero throughout training. The anti-correlation emerges within the first 25K steps, coinciding with the CIVO onset window (§D.8). This provides the representation-level smoking gun: trained gradient flow drives the embedding to encode dt=1 and dt=2 as maximally differentiated directions, creating the asymmetric latent space that underlies RSSM prediction bias.

D.8. dt-Embedding Differentiation Precedes Value Divergence

The preceding analysis established that trained dt-embeddings learn near-antipodal representations. Here we resolve the *temporal dynamics* of this differentiation across 11 experiments (∼47 checkpoints), testing whether embed-

ding differentiation is a consequence or precursor of CIVO.

Trained embeddings: monotonic relaxation. Across all trained configurations, $\cos(e_1, e_2)$ starts near -0.95 within 10K steps and relaxes monotonically toward ~ -0.65 (Table 24).

Cross-seed consistency at 25K steps is high: mean $\cos = -0.910 \pm 0.028$ (n=5 Walker seeds), confirming the relaxation pattern is robust across random seeds. Variance increases with training (-0.789 ± 0.076 at 75K), reflecting seed-dependent relaxation rates.

Temporal precedence over CIVO. Exp_066b provides simultaneous cosine similarity and CIVO diagnostic data at 5K resolution (Table 25).

Cosine relaxation begins at $\sim$20K (cos transitions from -0.948 at 15K to -0.928 at 25K), while the VE ratio first exceeds 0.7 at 55K. The $\sim$30–35K temporal lead establishes that embedding differentiation is an early training dynamic, not a consequence of CIVO-triggered feedback. The relaxation rate peaks at 25K–45K ($+0.024$/10K steps) before decelerating, consistent with the geometric intuition that large-angle adjustments in embedding space are cheaper early in training.

Embedding norms. Walker and Cheetah exhibit qualitatively different norm dynamics. Walker $e(\mathrm{dt}{=}1)$ norms grow from 0.23 to 0.50 ($2.2\times$) while $e(\mathrm{dt}{=}2)$ grows from 0.45 to 0.63 ($1.4\times$)—both dt values become more expressive over training. Cheetah norms remain nearly constant ($\leq 1.04\times$) despite continued cosine relaxation, indicating direction changes without magnitude change. Frozen embedding norms are an order of magnitude smaller ($\sim$0.06, constant).

Mechanistic interpretation: four-stage causal chain. The temporal ordering—embedding differentiation ($\sim$20K) $\rightarrow$ reward fidelity gap ($\sim$20K, §D.9) $\rightarrow$ value overestimation ($\sim$30K) $\rightarrow$ entropy rebound ($\sim$35K+)—reveals a four-stage causal chain supported by converging evidence from independent analyses:

1. **Embedding anti-correlation** (Figure 14): The trained dt-embedding encodes dt=1 and dt=2 as near-antipodal directions ($\cos \in [-0.65, -0.95]$), pushing the RSSM into qualitatively different latent regions per temporal scale.
2. **RSSM extrapolation failure** (Table 16): The anti-correlated representations cause environment-dependent reward fidelity gaps (Walker $1.63\times$; Cheetah: alternative pathway via prior drift without reward gap).
3. **Delusional critic** (Figure 10): The critic internalizes biased imagined returns, producing value estimates *negatively* correlated with actual returns ($r = -0.20$), rather than merely overestimated.
4. **Irreversible policy degeneration** (§D.10): The actor, trained on delusional critic gradients, eventually loses policy structure entirely (exp_097: entropy rebounds $+4.22$ nats from converged minimum).

The frozen embedding's constant near-zero cosine ($|\cos| < 0.2$) prevents Stage 1, explaining why freezing eliminates CIVO despite maintaining identical forward-pass architecture (§C.1).

D.9. Time-Series Fidelity–Value Analysis

To test the causal direction between reward fidelity degradation and value overestimation, we analyze exp_066 and exp_066b: two Walker Walk seeds (trained dt-embedding, fixed dt=2) evaluated at 12 dense checkpoints from 10K to 65K steps (Table 26).

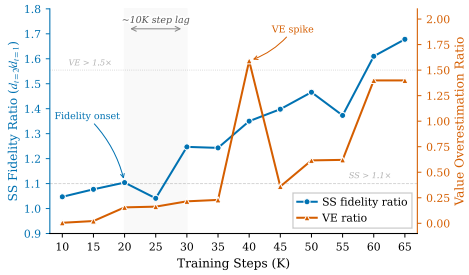


Figure 15. Temporal ordering of fidelity degradation and value overestimation (exp_066, seed 0). **Left axis**: single-step fidelity ratio (SS, blue) grows steadily from $1.05\times$ to $1.68\times$. **Right axis**: value-to-eval ratio (VE, red) remains near zero until 40K, then spikes. The $\sim$10K onset-to-onset lag supports the causal direction: fidelity gap $\rightarrow$ value overestimation. Seed 1 (exp_066b) replicates the pattern with a $\sim$9K temporal lead.

The dt=1 prediction MAE decreases monotonically ($0.090 \rightarrow 0.047$) as the world model improves, but dt=2 MAE plateaus at $\sim$0.08. The growing fidelity gap is therefore not caused by dt=2 predictions worsening, but by the model selectively improving at dt=1 while failing to transfer that improvement to dt=2—consistent with the trained dt-embedding learning maximally differentiated representations (§D.7) that prevent cross-scale generalization.

RSSM onset proxy (exp_066b). Exp_066b provides an independent proxy measurement of reward fidelity dynamics across the CIVO onset window. The reward prediction ratio ($\mathrm{MAE}_{\mathrm{dt}=2}/\mathrm{MAE}_{\mathrm{dt}=1}$) increases from $0.89\times$ at 20K to $1.80\times$ at 65K, confirming that the fidelity gap develops progressively and precedes CIVO onset—consistent with the checkpoint-level analysis above. The monotonic increase in the proxy confirms that fidelity degradation is a gradual process rather than a sudden phase transition.

Limitation. These results are limited to two seeds (n=2). While both seeds show consistent temporal ordering (leads of ≈10K and ≈9K steps), additional seeds would strengthen the causal claim.

D.10. Actor Entropy Is Not Diagnostic of CIVO

To test whether actor entropy can serve as an early collapse indicator, we compare entropy trajectories between collapsed and healthy configurations at matched training steps across five experiments (Table 27).

At the last comparable step (55K): collapsed mean entropy $= -5.02$ (average of -4.88, -5.17); healthy mean entropy $= -5.05$ (average of -4.77, -5.21, -5.18); difference $= 0.03$ nats—statistically negligible. The hypothesis that collapsed runs exhibit anomalous entropy rise (actor confusion from delusional critic gradients) is not supported during the CIVO onset window.

The fixed_dt2 experiment demonstrates this most clearly: returns peak at ∼580 (40K) then crash to 29 (72K), yet entropy remains stable at -5.1 throughout. The actor maintains a committed, low-entropy policy during the entire collapse—it does not become confused but rather remains confident while acting on incorrect value estimates. This is consistent with CIVO operating through critic value distortion rather than actor policy degradation. The late-stage entropy explosion documented in §D.1 ($+2.1$ nats/10K in EG v4, onset ∼35K steps after return collapse) occurs well beyond this window, confirming entropy rise as a lagging consequence rather than a concurrent or predictive signal.

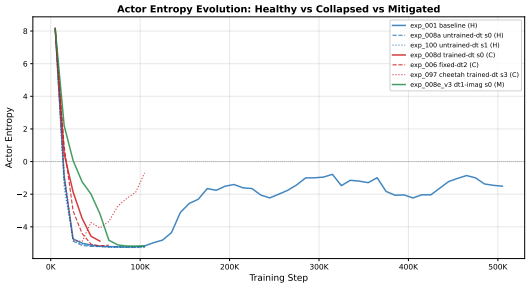


Figure 16. Actor entropy evolution across healthy (H, blue), collapsed (C, red), and mitigated (M, green) configurations. All runs converge to ∼−5 nats by 50K steps, rendering collapsed and healthy runs indistinguishable during the CIVO onset window. The baseline (exp_001, extended to 505K) shows adaptive entropy recovery to ∼−1.5 nats coinciding with continued performance improvement. Exp_097 (Cheetah, dotted red) exhibits CIVO-induced entropy rebound from -4.9 to -0.7, confirming irreversible policy degeneration (§D.1).

Entropy descent speed. Entropy descent rate correlates with temporal configuration complexity, not collapse fate. Simpler configurations (baseline dt=1, unexposed exp_008a) reach -4.7 by 25K, while configurations exposed to temporal variation (both collapsed and healthy) remain near -2.0 at 25K, reflecting the difficulty of policy commitment when training data spans multiple temporal scales.

CIVO-induced entropy rebound (exp_097). Cheetah Run exp_097 (onset-transient configuration) provides direct evidence of CIVO-induced irreversible policy degeneration. Actor entropy converges normally to -4.91 at 35K (comparable to healthy configurations at ∼-5.2), then *rebounds* to -0.69 at 105K—a $+4.22$ nat increase over 70K steps ($+0.60$ nats/10K), indicating complete strategy loss. Healthy and mitigated configurations (baseline, 008a, 008e_v3) converge monotonically to ∼-5.2 without rebound. This distinguishes CIVO-induced entropy rise from the adaptive entropy recovery in healthy runs (see below): in exp_097, the rebound coincides with persistent return degradation, confirming that the actor's policy structure has been irreversibly damaged by training on delusional critic gradients.

Long-horizon entropy recovery. The baseline (dt=1) run extended to 505K steps shows entropy recovering from -5.2 to ∼-1.5 between 115K and 295K steps, coinciding with continued performance improvement ($700 \rightarrow 950$ return). This adaptive entropy increase rules out entropy rise as an inherently negative signal; the actor maintains ∼950 return with entropy 3–4 nats above its minimum, indicating a healthy exploration–exploitation balance during extended training.

D.11. dt=1 Imagination as CIVO Mitigation

Restricting imagination to dt=1 while retaining the trained embedding (008e_v3) eliminates CIVO in all four tested seeds (NVE $\leq 0.8\times$): multi-step imagination at dt > 1 is the necessary trigger for the CIVO feedback loop. Three of four seeds achieve 64–94% of baseline (eval 597–877); the fourth (seed 0, eval 148) exhibits CIVO-unrelated learning stagnation. This mitigation preserves the world model's ability to distinguish temporal scales (unlike embedding removal) while eliminating the imagination-side bias that drives value overestimation (Figure 4).

D.12. Implications for Temporal Abstraction in World Models

CIVO identifies a structural incompatibility between flat temporal conditioning and imagination-based actor-critic training. Hierarchical approaches (THICK (Gumbsch et al., 2024), Hieros (Mattes et al., 2024), LS-Imagine (Li et al., 2025)) may avoid this by maintaining separate dynamics per temporal scale. Whether CIVO transfers to transformer-based world models (DreamerV4; Hafner et al. 2025b) depends on whether attention weights co-

adapt with temporal embeddings analogously. The CIVO mechanism requires three structural properties: (1) a temporal embedding that co-adapts with dynamics via gradient flow, (2) multi-step imagined rollouts used for critic training, and (3) a reward head that inherits latent biases from the shared representation. Transformer-based world models that share these properties may exhibit analogous vulnerability; architectures that decouple temporal conditioning from the reward prediction pathway would not.

D.13. Cross-Environment TCN Evaluation

Table 28 reports the full TCN evaluation across environments and noise magnitudes.

Cheetah Run σ=0.5: transient spike with permanent eval damage. Cheetah σ=0.5 exhibits a novel failure mode: a transient CIVO spike (NVE = 10.67× at 55K) followed by NVE recovery ($<$ 0.3× by 95K), but permanent evaluation damage (26% of baseline). The value signal self-corrects while the policy does not recover, suggesting that CIVO-induced representational damage during the spike persists within the observation window even when the critic subsequently recalibrates. This trajectory parallels the obs-decoder SGS seed 3 pattern (§D.3), where partial policy recovery coexists with persistent critic damage.

Cheetah Run σ=0.2: borderline prevention. At σ=0.2, Cheetah reaches NVE = 1.9× at 55K—below the 2.0× threshold but substantially higher than Walker's effective-dose NVE (0.64–0.91×). Although technically classified as CIVO-prevented by the NVE criterion, the eval crashed, indicating that the noise dose attenuates value overestimation just enough to stay below threshold while failing to prevent operational damage. This borderline case illustrates that the 2.0× NVE threshold captures the value-channel pathology but not all downstream failure modes. Increasing σ to 0.5 trades borderline prevention for a different failure mode (transient spike with permanent eval damage) rather than achieving robust mitigation.

PickCube-v1 σ=0.2: manipulation failure. PickCube reaches NVE = 3.71× at 52.5K with σ=0.2, indicating that TCN provides no protection in manipulation tasks. The combination of contact-gated dynamics and temporal conditioning may amplify the fidelity gap beyond what representation-level noise can attenuate.

Dose-response analysis. We observe σ=0.2 as a local optimum on Walker Walk, achieving complete CIVO suppression in 2 of 4 seeds. Seeds 2 and 3 both fail but via distinct CIVO onset dynamics: seed 2 shows gradual escalation (NVE = 2.11× at 95K, still rising, eval ∼5%), while seed 3 exhibits a phase-transition-like onset—NVE declined to 0.67× at 95K before spiking to 3.04× at 105K (eval 349.6 → 71.8 in 10K steps). When TCN fails to prevent CIVO onset, the resulting trajectory qualitatively resembles untreated CIVO in its terminal outcome—runaway NVE and eval collapse. The two failure modes (gradual escalation vs. sudden onset after apparent suppression) reinforce that TCN operates as a binary gate on CIVO onset rather than providing graded attenuation. Below the optimum, we observe σ=0.1 as insufficient (NVE = 1.02×—borderline, eval 55%); above it, σ=0.5 overdoses (oscillatory CIVO, eval $\leq$ 41%). Cross-seed variance at σ=0.2 (eval 5–91%) exceeds the cross-dose difference by a wide margin, reinforcing that seed effects dominate dose effects. TCN is the only tested intervention achieving complete CIVO suppression in any configuration, but its high seed sensitivity and failure to transfer across environments position it as a *diagnostic probe*—identifying the representation level as the relevant intervention target—rather than a deployable mitigation.

Cheetah Run "impossible dose" dilemma. Exp_030c (224K steps) reveals a qualitatively different dose-response on Cheetah. At σ=1.0, CIVO is eliminated (NVE≤0.30 throughout) and initial learning is *unimpaired*: peak eval reaches 372 at 35K (116% of baseline), demonstrating that σ=1.0 noise does not prevent policy acquisition. However, cumulative actor gradient interference produces progressive policy erosion—eval declines from 372 to 47 by 215K (15% of baseline)—while the critic remains well-calibrated (value_mean=12.4, NVE=0.26 at 215K). This establishes that *policy degradation* $\neq$ *value corruption*: accurate value signals are insufficient when the actor cannot optimize effectively in noise-perturbed imagination.

This "correct values, failed policy" pattern provides indirect evidence that CIVO is mechanistically distinct from standard overestimation. Double Q-learning and ensemble methods address critic estimation error, but at σ=1.0 the critic is accurate and the policy still collapses—a failure mode orthogonal to what these countermeasures target. The impossible dose dilemma scales with environment complexity: Cheetah's baseline CIVO severity (8–108× NVE) requires $\sigma \geq 1.0$ to suppress, but this exceeds the actor's gradient signal-to-noise tolerance. On Walker, a narrow effective window exists around σ=0.2; on Cheetah, the effective dose ($\sigma \geq 1.0$) exceeds the tolerable dose, and environments with higher CIVO severity will systematically lack a viable dose window.

E. DreamerV3 Hyperparameters

Table 29 lists the DreamerV3 hyperparameters used across all experiments. Unless otherwise noted, we use the default DreamerV3 configuration for continuous-control

tasks (`dmc_proprio`).

F. Online CIVO Detector Details

This appendix provides the full algorithm specification and validation results for the online CIVO detector described in §4.7.

F.1. Algorithm Specification

Algorithm 1 Online CIVO Detector. Monitors the value-to-evaluation ratio $r_t = \bar{V}_t / \bar{E}_t$ (not NVE) for superlinear growth indicative of CIVO onset.

Require: Stream of $(\bar{V}_t, \bar{E}_t)$ at each checkpoint t
$\tau_{\text{mag}} \leftarrow 0.49, \quad \tau_{\text{vel}} \leftarrow 0.1, \quad \varepsilon \leftarrow 10^{-6}$
Initialize $r_{\text{prev}} \leftarrow 0$, $\Delta r_{\text{prev}} \leftarrow 0$, $\mathcal{H} \leftarrow []$
for each checkpoint t **do**
  $r_t \leftarrow \bar{V}_t \,/\, \max(\bar{E}_t, \varepsilon)$
  $\Delta r_t \leftarrow r_t - r_{\text{prev}}$
  $c_{\text{mag}} \leftarrow (r_t \geq \tau_{\text{mag}})$
  $c_{\text{vel}} \leftarrow (\Delta r_t > \tau_{\text{vel}})$
  $c_{\text{acc}} \leftarrow (\Delta r_{\text{prev}} > \tau_{\text{vel}}) \wedge (\Delta r_t > \Delta r_{\text{prev}})$
  $c_{\text{dec}} \leftarrow \left|\{i : \mathcal{H}[i] < \mathcal{H}[i{-}1],\ i \in \text{last } 3\}\right| \geq 2$
  if $c_{\text{mag}} \wedge c_{\text{vel}} \wedge c_{\text{acc}} \wedge c_{\text{dec}}$ **then**
    Alert: CIVO onset at step t
  end if
  $r_{\text{prev}} \leftarrow r_t, \quad \Delta r_{\text{prev}} \leftarrow \Delta r_t$
  Append $\bar{E}_t$ to $\mathcal{H}$
end for

The four conditions capture complementary signals: (1) *Magnitude* ($r_t \geq 0.49$) confirms that the value-to-evaluation ratio has reached a level indicative of overestimation; (2) *Velocity* ($\Delta r_t > 0.1$) rules out slow drift or plateau behavior; (3) *Acceleration* ($\Delta r_t > \Delta r_{t-1}$, both > 0.1) filters oscillatory paths; (4) *Evaluation decline* (≥ 2 of 3 recent checkpoints decline) confirms that overestimation coincides with actual performance degradation. A full comparison of window sizes appears in §B.

Sensitivity analysis over $\tau_{\text{mag}} \in [0.30, 0.70]$ confirms that $\tau_{\text{mag}} = 0.49$ maximizes Youden's $J = 0.714$; raising the threshold to 0.50 loses one true positive (seed 3 triggers at $r_t = 0.497$) with no reduction in false positives. The velocity threshold $\tau_{\text{vel}} = 0.1$ is stable across $[0.05, 0.18]$ (TPR remains $5/5$).

F.2. Validation Results

The detector identifies all five CIVO-positive runs (TPR = 5/5). Two are detected predictively (008d seeds 0 and 3, both at 45K). Two false positives occur among nine negative cases (FPR = 2/9, precision 71%, $F_1 = 0.83$). Run 009c develops genuine late-stage value overestimation unrelated to CIVO; run 009a triggers during a transient eval dip at 85K before recovery to 966. Early alerts (before 70K) carry higher diagnostic confidence.

F.3. Cross-Simulator Specificity (PickCube)

Exp_019b applies the CIVO-trigger configuration to PickCube-v1 in ManiSkill3, a dense-reward manipulation task. The V/E ratio exhibits an oscillatory pattern rather than monotonic divergence: it spikes to $r_t = 5.66$ at 52.5K, then deflates to 1.38 by 72.5K. This trajectory prevents the acceleration condition from co-occurring with evaluation decline at any single checkpoint. Across all 14 checkpoints through 132.5K, the detector correctly remains silent (TN), confirming that the acceleration filter distinguishes transient spikes from CIVO's sustained monotonic divergence. A frozen-embedding control (exp_054) confirms this classification: frozen PickCube-v1 exhibits a comparable burst (peak $8.81\times$ at 62.5K) that resolves within a single evaluation cycle ($0.55\times$ at 72.5K), establishing environment-specific origin independent of dt-embedding training.

F.4. Detector Limitations

Four constraints bound the current detector. First, the three-checkpoint window cannot distinguish transient dips from sustained collapse. Second, the acceleration condition assumes monotonically increasing overestimation during onset. Third, the detector identifies a *pattern* rather than a *mechanism*: run 009c shows that non-CIVO pathologies produce the same signature. Fourth, threshold values were calibrated on fourteen runs across three environments; generalization to different reward scales requires further validation.

G. Dual-Critic Mechanism Ablation

G.1. 2×2 Factorial Design

We ablate the critic architecture in a 2×2 factorial: {single, dual} $\times$ {narrow ($\sim$1.2M), wide ($\sim$2.4M)} parameters. Table 31 reports all configurations.

G.2. Critic Convergence ($v_1 \approx v_2$)

Across all dual-critic configurations and environments, the two critics converge to near-identical value functions. On Walker Walk, Pearson $\rho > 0.999$ between v_1 and v_2 outputs; on Cheetah Run and PickCube-v1, the gap is < 0.2 (exact match within numerical precision). Per-sample pessimism $\mathbb{E}[\max(v_1, v_2) - \min(v_1, v_2)]$ is 3–4% of the theoretical bound for independent critics.

This convergence occurs because both critics train on the same RSSM-generated latent features. Unlike model-free Double Q-learning, where each critic sees different mini-

batches of real transitions and develops distinct function approximation errors, pure-imagination critics share identical inputs from the world model. The min operation therefore cannot provide the pessimism bias that makes Double Q effective in model-free settings.

G.3. Recovery vs. Prevention Mechanism

The 2×2 factorial reveals two separable dimensions:

Recovery (min-dominated). The min operation is the primary factor enabling recovery from CIVO. Both narrow dual-critic seeds experience CIVO and recover fully: exp_035 (s0, $\mathrm{nve} = 2.75\times$, eval $33.8 \rightarrow 697.8$, recovery ratio $20.6\times$) and exp_039 (s1, $\mathrm{nve} = 2.19\times$, eval $35.9 \rightarrow 592.7$, recovery ratio $16.5\times$). Both outperform wide single-critic (exp_033, 2.4M params, eval 407.9) despite $2\times$ fewer parameters. The min operation reduces both peak CIVO severity (2.19–$2.75\times$ vs. $4.75\times$ for the comparable single-critic) and enables consistent full recovery across seeds.

Prevention (min + capacity synergy). CIVO prevention (NVE remaining below the $2.0\times$ threshold) requires both the min operation and increased capacity: only the wide dual-critic (exp_032, 3 seeds) achieves this. The mechanism for this synergy is an open question; we hypothesize that increased capacity improves value calibration quality while the min operation constrains the overestimation feedback loop during early training when the two critics briefly diverge.

Cross-environment limits. On Cheetah Run, the wide dual-critic (exp_034) reduces NVE from $8.27\times$ (untreated baseline) to $1.75\times$ ($\geq 4\times$ reduction) but cannot prevent CIVO onset, and recovery reaches only 31% of the pre-CIVO peak. This confirms that the min + capacity synergy has environment-dependent limits: in high-CIVO-severity environments, architecture-level interventions attenuate but do not eliminate the pathology.

Table 3. Complete experimental results. NVE reports peak nve in the 35K+ window (§3.4). CIVO-positive seeds are shaded; † marks self-correcting NVE without policy recovery.

Exp	Config	dt_emb	Sub	H	Eval@final	NVE	Outcome
001	Baseline (dt=1, s1)	—	—	15	915.7 @105K	1.00×	Healthy
001 s2	Baseline (dt=1, s2)	—	—	15	443.5 @105K	—[e]	Healthy
006	Fixed dt=2	trn	ON	15	132.9 @65K	4.83×	Collapsed
007	Fixed dt=4	trn	ON	15	152.5 @55K	—	Collapsed
005	Learned gate	adaptive	OFF	15	98.8 @145K	—	Collapsed
008d	Factorial (s1)	trn	ON	8	82.8 @55K[♡]	4.12×	Full collapse
009b	Factorial (s2)	trn	ON	8	267.9 @105K	1.62×	Transient CIVO
008d_s3	Factorial (s3)	trn	ON	8	268[◇] @105K	1.53×	Late-onset collapse
008d_s4	Factorial (s4)	trn	ON	8	63.6 @105K	4.62×	Full collapse
059	Factorial (s5)	trn	ON	8	95.8 @105K	4.72×	Collapse[†]
068	Factorial repro (s1)	trn	ON	8	39.1 @105K	4.12×	Full collapse
008f	No dt_emb (s0)	—	ON	8	139.5 @105K[l]	1.69×[l]	Healthy[l]
— s1	No dt_emb (s1)	—	ON	8	468.8 @105K	0.88×	Healthy
008f s4	No dt_emb (s4)	—	ON	8	289.3 @105K	0.76×	Healthy
009c s3	No dt_emb (s3)	—	ON	8	311.9 @105K	1.04×	Healthy
008a	Unexposed emb	exp[a]	OFF	15	777.7 @105K	0.93×	Healthy
008a_s1	Unexposed emb	exp[a]	OFF	15	630.9 @105K	0.83×	Healthy
008a_s2	Unexposed emb	exp[a]	OFF	15	899.4 @105K	0.89×	Healthy
008a_s3	Unexposed emb	exp[a]	OFF	15	855.9 @105K	—	Healthy
015b	Frozen emb	frz[b]	ON	8	204.2 @105K	1.45×	Grad. starvation
022b	Frozen emb + Gumbel gate	frz[b]	ON	8	300.7 @105K	0.63×	No CIVO
042	Frozen emb (dedicated)	frz[b]	ON	8	612.0 @105K	0.19×	Healthy
047	Frozen emb (s1)	frz[b]	ON	8	144.2 @215K	1.71×	Transient (self-corr.)
057	Frozen emb (s2)	frz[b]	ON	8	123.9 @215K	1.88×	Transient (self-corr.)
067	Frozen emb (s0)	frz[b]	ON	8	281.5 @105K	0.19×	Healthy
018 s1	Full-SGS	trn[‡]	ON	8	448.3 @105K	1.66×[*]	Partial mitig. (CIVO+)
018 s2	Full-SGS	trn[‡]	ON	8	40.8 @105K	—	Collapsed
008e_v3 s0	Mitigation	trn	ON	15	148.0 @105K	0.80×	No CIVO[j]
008e_v3 s1	Mitigation	trn	ON	15	876.7 @105K	0.60×	Healthy
008e_v3 s2	Mitigation	trn	ON	15	597.1 @105K	0.50×	Healthy
008e_v3 s5	Mitigation	trn	ON	15	853.1 @105K	0.61×	Healthy
Cross-environment validation							
010b	Cheetah Run	trn	ON	8	51.7 @105K	8.27×	Collapsed
010b_s0	Cheetah Run	trn	ON	8	0.83 @105K	108.52×	Full collapse
021	Cheetah Run	trn[‡]	ON	8	221.5 @105K	9.94×[d]	Transient CIVO (self-corr.)
024b s0	Cheetah Run	—	ON	8	96.5 @105K	<0.20×	Healthy
024b s1	Cheetah Run	—	ON	8	77.3 @105K	0.10×	Healthy
092 s1	Cheetah Run	—	ON	8	77.3 @105K	<1.0×[n]	Healthy
093 s2	Cheetah Run	—	ON	8	151.2 @105K	<1.0×[n]	Healthy
008a_ch s0	Cheetah Run[k]	exp[a]	OFF	15	366.2 @105K	0.45×	Healthy
008a_ch s1	Cheetah Run[k]	exp[a]	OFF	15	281.2 @105K	0.58×	Healthy
077	Cheetah baseline (s0)	—	—	15	448.1 @105K	1.00×	Healthy
083	Cheetah baseline (s1)	—	—	15	444.0 @105K	1.00×	Healthy
078	Cheetah trigger (s0)	trn	ON	8	119.5 @105K	4.10×	Onset (self-corr.)
097 s3	Cheetah trigger (s3)	trn	ON	8	61.7 @105K	≥2.0×[o]	Onset (self-corr.)
019b	PickCube-v1	trn	ON	8	26.64 @200K	3.78×[c]	Transient onset
054	PickCube-v1	frz[b]	ON	8	—	8.81×[h]	Env. transient (ctrl)
049	StackCube-v1	trn	ON	8	34.9 @212K	5.05×[f]	Env. transient
052	StackCube-v1	trn	ON	8	—	4.67×[f]	Env. transient
051	StackCube-v1	frz[b]	ON	8	30.7 @155K	4.30×[f]	Env. transient (ctrl)
050	PegInsertionSide-v1	trn	ON	8	20.86 @212K	10.05×[g]	Env. dynamics
053	PegInsertionSide-v1	frz[b]	ON	8	5.89	8.10×[g]	Env. dynamics (ctrl)
055	PegInsertionSide-v1	trn	ON	8	11.44 @212K	3.08×[g]	Env. dynamics
056	PegInsertionSide-v1	frz[b]	ON	8	4.37 @212K	11.32×[g]	Env. dynamics (ctrl)
046	Cheetah Run	frz[b]	ON	8	150.5 @215K	1.35×	No CIVO
058	Cheetah Run	frz[b]	ON	8	80.46 @215K	0.78×	No CIVO
Cartpole Swingup							
060	Cartpole baseline (3 seeds)	—	—	15	557.0 @105K[i]	—	Healthy
061	Cartpole trigger (3 seeds)	trn	ON	8	236.3 @105K[i]	—	No CIVO
062	Cartpole control (3 seeds)	—	ON	8	244.8 @105K[i]	—	No CIVO

trn = trained (d=32, subsample ON); trn[‡] = full-pathway SGS (all non-KL gradients detached); exp[a] = trained, subsample OFF; frz[b] = frozen (d=32, `requires_grad=False`); — = removed (d=0).

[*] Seed 1 only: peak NVE with transient exceedances (1.60× at 35K, 1.66× at 95K); self-correcting, no sustained escalation. Seed 2 collapsed.

[◇] Transitional; train returns still declining (peak 470 to 139 over final 8K steps). Collapse had not reached equilibrium at 105K.

[†] NVE self-corrected ($4.72\times \rightarrow 1.22\times$) but policy did not recover (eval 10.5% of baseline).

[c] NVE normalized against exp_019a_v2 (vanilla DreamerV3 on PickCube-v1).

[d] Peak NVE $9.94\times$ at 45K (transient, self-correcting to $<1\times$ by 55K); final eval 221.5.

[e] Not used for NVE normalization; reported for completeness. Seed 1 (eval 915.7) is the NVE reference baseline.

[f] StackCube transient: frozen control (exp_051, peak $4.30\times$ at 42.5K) matches trained (exp_049/052), self-correcting by 82.5K; environment-specific, not dt-embedding-driven.

[g] PegInsertion environment dynamics: seed 0 trained peaks at $10.05\times$ (82.5K); seed 1 trained (exp_055) peaks at $3.08\times$ (92.5K), final $0.50\times$ at 212.5K; seed 1 frozen (exp_056) peaks at $11.32\times$ (82.5K), final $1.19\times$ at 212.5K. Frozen final exceeds trained final; ordering is seed-dependent, ruling out dt-embedding causation.

[h] PickCube environment transient: frozen control peak $8.81\times$ at 62.5K, self-correcting to $0.55\times$ at 72.5K; matches trained (exp_019b) trajectory.

Table 4. Cross-domain CIVO reproducibility. "dt-emb-dep." indicates trained peaks consistently exceed frozen at matched timepoints across seeds.

Domain	**Environment**	**Trained**	**Frz ctrl**	**dt-emb-dep. CIVO**
Locomotion	Walker Walk	2/3	0/3	Yes (NVE 4.12–4.13×)
Locomotion	Cheetah Run	3/4	0/2	Yes (NVE 4.10–138.7×)
Manipulation	StackCube-v1	0/2	0/1	No (frozen≈trained)
Manipulation	PegInsertionSide	0/2	0/2	No (seed-dep. ordering)
Manipulation	PickCube-v1	0/1	0/1	No (frozen spike similar)
Locomotion	Cartpole Swingup	0/3	0/3	No (single-joint)

Table 5. Cross-environment CIVO comparison. **Trigger**: trained dt_emb + subsample ON + H=8; **Control**: dt_embed_dim=0 (no dt_emb) + subsample ON + H=8. NVE Peak reports the maximum Normalized Value Excess across all seeds. CIVO is flagged when any seed exceeds NVE $\geq 2.0\times$. Per-row seed lists in footnotes; sample std throughout.

Environment	Configuration	Eval Return	NVE Peak	CIVO
Walker Walk	Baseline (Δt=1)	787.7 ± 230.8^{a}	—	—
	Trigger (trained dt_emb)	146.9 ± 112.3^{b}	$4.72\times$	✓ (2/3)
	Control (no dt_emb)	302.4 ± 134.8^{c}	$1.69\times^{*}$	✗ (0/4)
Cheetah Run	Baseline (Δt=1)	446.1 ± 2.9^{d}	—	—
	Trigger (trained dt_emb)	77.6 ± 36.6^{e}	$8.27\times$	✓ (3/4)
	Control (no dt_emb)	100.6 ± 34.9^{f}	$0.20\times$	✗ (0/4)
Cartpole Swingup	Baseline (Δt=1)	673.3 ± 142.0	—	—
	Trigger (trained dt_emb)	236.3 ± 29.7	$0.06\times$	✗ (0/3)
	Control (no dt_emb)	244.8 ± 12.7	$\leq 0.1\times$	✗ (0/3)

[a] N=4: exp_001 s1 (915.7), s2 (443.5), s3 (869.0), s4 (922.8); s3/s4 from cross-environment evaluation runs. [b] Eval: N=5 at 105K (009b, 008d_s3, 008d_s4, 059, 068). CIVO evaluated on N=3 seeds with full NVE pipeline (008d s0, 009e s1, 009b s2); 2/3 exceed 2.0×. [c] N=4: 008f s0 (139.5), s1 (468.8), s4 (289.3), 009c s3 (311.9). [d] N=2: exp_077 s0 (448.1), exp_083 s1 (444.0); consistent with Table 3. [e] N=3: exp_010b s1 (51.7), exp_078 s0 (119.5), exp_097 s3 (61.7). exp_010b s0 (eval=0.8, NVE=108.5×) excluded as outlier. [f] N=4: exp_024b s0 (96.5), s1 (77.3), exp_092 s1 (77.3), exp_093 s2 (151.2). NVE Peak from N=2 seeds with computed NVE. [*] 008f s0 NVE 1.69× reflects endpoint eval dip (95K: 342.9 → 105K: 139.5), not value overestimation; classified non-CIVO.

Table 6. Per-experiment NVE values (35K+ steps) and ground truth labels. + = positive (CIVO), − = negative. Trans. = transient CIVO with self-correction.

Experiment	NVE	Mechanism	B1	B2
006	4.83	sustained CIVO	+	+
059	4.72	self-correcting CIVO	−	+
008d	4.12	sustained CIVO	+	+
009b	1.62	transient CIVO	−	+
001	1.00	non-CIVO	−	−
008f	1.69[l]	non-CIVO	−	−
008a	0.93	non-CIVO	−	−
008e_v3_s0	0.60	non-CIVO	−	−
008e_v3_s1	0.60	non-CIVO	−	−
008e_v3_s5	0.60	non-CIVO	−	−
008e_v3_s2	0.50	non-CIVO	−	−

Table 7. **Task B1**: Sustained CIVO detection sensitivity analysis ($n_{\text{pos}} = 2$, $n_{\text{neg}} = 9$, $n = 11$, AUROC $= 0.944$). All confidence intervals are two-sided Clopper-Pearson exact binomial ($\alpha = 0.05$). †Optimal threshold in $(1.62, 4.12]$; exp_059 (NVE=$4.72\times$, B1-negative) is the sole false positive.

Threshold	TP	FP	FN	TN	Accuracy [95% CI]	Precision	Recall	F1
$\geq 1.0\times$	2	3	0	6	72.7% [39.0, 94.0]	40.0%	100%	57.1%
$\geq 1.2\times$	2	2	0	7	81.8% [48.2, 97.7]	50.0%	100%	66.7%
$\geq 1.5\times$	2	2	0	7	81.8% [48.2, 97.7]	50.0%	100%	66.7%
$\geq 1.7\times^{\dagger}$	2	1	0	8	90.9% [58.7, 99.8]	66.7%	100%	80.0%
$\geq 2.0\times$	2	1	0	8	90.9% [58.7, 99.8]	66.7%	100%	80.0%
$\geq 3.9\times$	2	1	0	8	90.9% [58.7, 99.8]	66.7%	100%	80.0%

Table 8. **Task B2**: Any CIVO detection sensitivity analysis ($n_{\text{pos}} = 4$, $n_{\text{neg}} = 7$, $n = 11$, AUROC $= 1.000$). Perfect rank separation: all CIVO-positive experiments have NVE $> 1.5\times$ and all negatives have NVE $\leq 1.0\times$. The $1.5\times$ threshold (highlighted) achieves perfect classification.

Threshold	TP	FP	FN	TN	Accuracy [95% CI]	Precision	Recall	F1
$\geq 1.0\times$	4	1	0	6	90.9% [58.7, 99.8]	80.0%	100%	88.9%
$\geq 1.2\times$	4	0	0	7	100.0% [71.5, 100.0]	100%	100%	100%
$\geq 1.5\times$	4	0	0	7	100.0% [71.5, 100.0]	100%	100%	100%
$\geq 1.7\times$	3	0	1	7	90.9% [58.7, 99.8]	100%	75.0%	85.7%
$\geq 2.0\times$	3	0	1	7	90.9% [58.7, 99.8]	100%	75.0%	85.7%
$\geq 3.9\times$	3	0	1	7	90.9% [58.7, 99.8]	100%	75.0%	85.7%

Table 9. **Task B2 (full evaluation set)**: Any CIVO detection across two environments ($n_{\text{pos}} = 8$, $n_{\text{neg}} = 9$, $n = 17$, AUROC $= 1.000$). Includes Walker Walk (15 configurations) and Cheetah Run (2 configurations). Perfect rank separation; all CIVO-positive NVE $\geq 1.53\times$, all negatives $\leq 1.04\times$.

Threshold	TP	FP	FN	TN	Accuracy [95% CI]	Precision	Recall	F1
$\geq 1.0\times$	8	2	0	7	88.2% [63.6, 98.5]	80.0%	100%	88.9%
$\geq 1.2\times$	8	0	0	9	100.0% [80.5, 100.0]	100%	100%	100%
$\geq 1.5\times$	8	0	0	9	100.0% [80.5, 100.0]	100%	100%	100%
$\geq 1.7\times$	7	0	1	9	94.1% [71.3, 99.9]	100%	87.5%	93.3%
$\geq 2.0\times$	7	0	1	9	94.1% [71.3, 99.9]	100%	87.5%	93.3%
$\geq 3.9\times$	6	0	2	9	88.2% [63.6, 98.5]	100%	75.0%	85.7%

Table 10. Walker Walk baseline seeds at 105K and 505K steps.

Seed	Eval@105K	Eval@505K	ve@65K	ve@105K
s1	915.7	970.4	0.563	0.656
s2	443.5	945.7	0.898	0.742
s3	869.0	947.7	—	0.664
s4	922.8	938.9	—	0.624

Table 11. Leave-one-environment-out cross-validation of NVE threshold $\theta = 2.0\times$. For each fold, the threshold is validated on an environment whose data was *not* used to select θ. The optimal threshold range on each training set is reported; $\theta = 2.0$ falls within the optimal range in all three folds.

Held-out	N_{train}	N_{test}	θ_{opt} range	Test Accuracy	Test AUROC
PickCube	20	2	$(1.45, 1.53]$	2/2 (1.000)	1.000
Cheetah	17	5	$(1.45, 1.53]$	5/5 (1.000)	1.000
Walker	7	15	$(1.00, 3.78)$	15/15 (1.000)	1.000
Mean	—	—	—	**1.000**	**1.000**

Table 12. dt-embedding gradient conditions on Walker Walk. Any gradient pathway updating e_{dt} is sufficient to induce CIVO; freezing or removing the embedding dramatically reduces severity.

Condition	Grad flow	Exp(s)	NVE peak	Eval	CIVO
			Multi-rate data (ON)		Single-rate data (OFF)
Trained	all	008d, 059	1.53–4.72×	64–268	3/5
			exp_008d: **CIVO**		exp_008a: healthy
Reward detach	obs+KL only	044	4.45×	44→538†	1/1
			NVE 4.12×, collapse		NVE 0.93×
Full SGS	KL only	018	1.66×	448	1/1*
Frozen	none	042, 015b, 047, 057, 084, 085	0.19–1.88×‡	204–821	0/6
			exp_015b: grad. starvation		—
Removed	N/A	008f	0.76–1.69×[l]	140–469	0/4
			peak NVE 1.45× (transient)		
Removed e_{dt}		exp_008f: no CIVO[l]			—
		NVE ≤1.69×[l]			

*Full-SGS seed 1 (NVE 1.66×); seed 2 collapsed.
†Self-corrected at ∼175K (eval 44→538, 59% baseline; 224K run). ‡No frozen seeds exceed 2.0×; two show transient spikes (exp_047: 1.71×; exp_057: 1.88×) that self-correct within 20K steps; the remaining 4 seeds stay below 0.53×.

Table 13. Three-way causal ablation (Walker Walk, seed=1). Only the dt-embedding treatment varies; all other configuration is identical. Trained embedding triggers CIVO; frozen and removed do not.

Exp	Embedding	NVE peak	Eval final	CIVO
068	Trained (d=32)	6.57×	39.1	Yes (alert @85K)
067	Frozen (d=32, `req_grad=F`)	0.40×	281.5	No
069	Removed (d=0)	0.66×	237.8	No

Table 14. Factorial design. Each row specifies one experimental configuration. **Emb**: dt-embedding mode. **Sub**: temporal subsampling. **H**: imagination horizon. **Imag** dt: temporal resolution used during imagination.

ID	Purpose	Emb	Sub	H	Imag dt
001	Reference baseline	—	—	15	1
006	Fixed dt=2	trn	on	15	2
007	Fixed dt=4	trn	on	15	4
008a	Unexposed emb ctrl	exp[a]	off	15	2
008c	Horizon control	exp[a]	off	8	2
008d	Collapse condition	trn	on	8	2
008e_v3	dt=1 imag control	trn	on	15	1
008f	Causal ablation	—	on	8	2
015b	Frozen-emb ctrl	frz[b]	on	8	2
016e	Stop-grad (reward)	trn†	on	8	2
018	Full-SGS	trn‡	on	8	2
005	Learned gate	adaptive	off	15	learned

trn = trained (d=32, exposed to dt>1 data via subsample ON); exp = trained (d=32) but never exposed to dt>1 data (subsample OFF); frz[b] = frozen random projection (d=32, `requires_grad=False`); — = removed (d=0).
† `reward_dt_emb_detach=True`; ‡ all non-KL gradients to `dt_emb` detached.

Table 15. Causal isolation: e_{dt} configuration × multi-rate data exposure. The combination of trained embedding and multi-rate exposure is the sole predictor of CIVO in tested configurations. Neither factor alone is sufficient.

Table 16. RSSM fidelity at dt=1 vs. dt=2 (v3 standardized). The reward prediction gap is environment-dependent: Walker trained embedding produces a 1.63× reward gap, while Cheetah shows no gap (1.00×) despite exhibiting CIVO.

Env	Config	obs_MSE dt=1	obs_MSE dt=2	obs_MSE Gap	reward_MAE dt=1	reward_MAE dt=2	reward_MAE Gap
Walker	Baseline	13.49	27.50	2.04×	0.0102	0.0159	1.56×
	Trained (008d)	25.71	34.60	1.35×	0.0530	0.0865	1.63×
	No emb (008f)	8.01	22.04	2.75×	0.7469	0.7526	1.01×
Cheetah	Trained	5.26	4.90	0.93×	0.1410	0.1405	1.00×

Table 17. Signed reward error (predicted − actual) on posterior-conditioned RSSM predictions. Negative values indicate under-prediction. The trained-embedding model underpredicts at dt=2, ruling out observation-level reward overprediction as the CIVO mechanism.

Config	dt	Mean SE	Std	%Over	Actual	Pred
Baseline	1	+0.001	0.023	39.6%	1.893	1.894
008d (trained)	1	+0.018	0.072	61.8%	0.361	0.379
008d (trained)	2	**−0.599**	0.722	8.0%	0.721	0.121
008f (no emb)	1	−0.710	0.704	14.0%	0.921	0.211
008f (no emb)	2	−1.645	1.485	5.0%	1.840	0.195

Table 18. Cosine similarity between $e_{\mathrm{dt}=1}$ and $e_{\mathrm{dt}=2}$ vectors. Trained embeddings converge to near-opposite representations; frozen embeddings remain near-orthogonal throughout.

Config	Env	cos@25K	cos@75K	Status
008d s3 (trained)	Walker	−0.87	−0.70	CIVO
059 s5 (trained)	Walker	−0.94	−0.86	CIVO
010b s0 (trained)	Cheetah	−0.90	−0.84	CIVO
057 s2 (frozen)	Walker	−0.15	−0.15	Healthy
058 s1 (frozen)	Cheetah	+0.07	+0.07	Healthy

Table 19. RSSM fidelity analysis on Walker Walk (training-time posterior-conditioned, pre-v3 standardization). Reward prediction gap magnitude varies across checkpoints and measurement protocols: this posterior-conditioned analysis yields 3.11×, while v3 standardized analysis (Table 16) yields 1.63×. The qualitative pattern—trained embedding amplifies reward gap while improving observation reconstruction—is consistent across both.

	obs_MSE			reward_MAE		
Config	dt=1	dt=2	Gap	dt=1	dt=2	Gap
Baseline (009d)	16.08	23.49	1.46×	0.045	0.059	1.30×
Collapsed (008d)	9.96	13.21	1.33×	0.057	0.178	3.11×
Frozen (057)	11.14	14.95	1.34×	0.068	0.133	1.96×

Table 20. Critic value predictions vs. actual evaluation returns. The CIVO model's critic is well-calibrated to its λ-return targets ($V(s) \approx$ target) but both V and targets are disconnected from reality (VE=3.03×).

Config	$V(s)$	Eval	VE ratio	$V-$target
Baseline (001)	665.9	938.9	0.71×	+0.001
CIVO (008d)	192.7	63.6	**3.03**×	+0.586
No emb (008f)	116.5	306.5	0.38×	+0.316

Table 21. Gradient contributions to the temporal embedding $e_{\Delta t}$ by loss component, measured via `torch.autograd.grad` with training configuration `gate_fixed_dt=2`. Gradient norms are ℓ_2 norms of $\nabla_\theta \mathcal{L}_i$ for each loss $\mathcal{L}_i$.

	@55K		@112K	
Loss component	Share	(%)	Share	(%)
Reward head	—	49.0	—	38.1
KL dynamics	—	34.7	—	40.4
Observation decoders	—	16.3	—	21.5

Table 22. Post-CIVO outcomes for all six runs exceeding the nve $\geq 2.0\times$ diagnostic threshold across two environments. Peak/Final NVE measured in the 35K+ window.

Env	Seed	Peak	Final	Eval	Outcome
Walker	s1 (008d)	4.12×	4.12×	82.8	Full collapse
Walker	s4 (008d_s4)	4.62×	4.62×	63.6	Full collapse
Walker	s5 (059)	4.72×	1.22×	95.8	Collapse†
Cheetah	s1 (010b)	8.27×	8.27×	51.7	Full collapse
Cheetah	s0 (078)	4.10×	0.29×	119.5	Self-correcting‡
Cheetah	s0 (010b_s0)	108.52×	108.52×	0.83	Full collapse

Two sub-threshold runs (009b, peak 1.62×; 008d_s3, peak 1.53×) are omitted. † NVE self-corrected (4.72× → 1.22×) but policy did not recover (eval 10.5% of baseline). ‡ Peak NVE 4.10× at 75K; eval 213.7→26.1→119.5 (26.8% of 2-seed baseline mean, 446.1).

Table 23. Failure and degradation mode taxonomy. Each mode is characterized by its NVE trajectory, entropy behavior, and primary mechanism.

Mode	Representative	NVE	Entropy	Mechani
Monotonic CIVO	008d, 008d_s4, 010b_s0	↑ sustained	↓	Reward b
Late-onset CIVO	008d_s3	↑ delayed	↓	Same, del
Transient CIVO	009b	Peak → ↓	Stable	Self-corre
Grad. starvation	015b	Oscillates, decays	Stable	Frozen en
SGS-resistant	018_s2	↑	↓	Forward-p
SGS-paradoxical	016e	↑ sustained	↓	Reward S

Table 24. Cosine similarity evolution between $e(\mathrm{dt}=1)$ and $e(\mathrm{dt}=2)$ for trained dt-embeddings. Relaxation is monotonic but never reaches zero; frozen embeddings remain constant near zero throughout training.

Env	Experiment	10K	25K	65K	75K	302K
Walker	exp_066b (trained)	−0.947	−0.928	−0.847	—	—
	exp_045 (extension)	—	—	—	—	−0.668
Cheetah	exp_010b→048	—	−0.904	—	−0.841	−0.639
Frozen controls:						
Walker	exp_057 (frozen)	−0.150 (constant, 25K–225K)				
Cheetah	exp_058 (frozen)	+0.069 (constant, 25K–225K)				

Table 25. Cosine similarity relaxation and CIVO diagnostic for exp_066b. Embedding relaxation onset (∼20K) precedes CIVO alert (55K, VE ratio > 0.7) by ∼30–35K steps.

Step	$\cos(e_1, e_2)$	VE ratio	CIVO alert
10K	−0.947	0.012	—
15K	−0.948	0.033	—
25K	−0.928	0.193	—
35K	−0.904	0.269	—
45K	−0.880	0.358	—
55K	−0.861	0.710	**Alert**
65K	−0.847	0.724	—

Table 26. Time-series fidelity and value metrics (exp_066, seed 0). SS = single-step MAE ratio (dt=2/dt=1); VE = value-to-eval ratio. Fidelity degradation (SS $>$ 1.1× at 20K) precedes value overestimation onset (VE begins rising at ∼30K) by ∼10K steps. Seed 1 (exp_066b) confirms the pattern: fidelity gap onset ∼15K → VE onset ∼24K (lead ≈9K steps; VE peak =0.72).

Step	SS ratio	OL ratio	VE ratio	$\text{MAE}_{\text{dt=1}}$	$\text{MAE}_{\text{dt=2}}$	Return
10K	1.05	0.94	0.01	0.090	0.094	43.0
15K	1.08	1.03	0.02	0.085	0.092	187.8
20K	1.10	1.09	0.16	0.078	0.086	148.7
25K	1.04	1.22	0.16	0.073	0.076	142.1
30K	1.25	1.45	0.22	0.068	0.085	148.3
35K	1.24	1.51	0.23	0.062	0.077	275.4
40K	1.35	1.38	**1.59**	0.058	0.078	202.3
45K	1.40	1.38	0.36	0.055	0.077	107.1
50K	1.47	1.56	0.62	0.052	0.076	178.4
55K	1.37	1.52	0.62	0.050	0.069	60.8
60K	1.61	1.35	1.40	0.049	0.079	125.3
65K	1.68	1.38	1.40	0.047	0.079	94.4

Table 27. Actor entropy comparison at matched time points. Collapsed and healthy configurations exhibit indistinguishable entropy at 55K steps (difference $=$ 0.03 nats), despite collapsed runs showing severe return degradation.

Group	Experiment	5K	25K	45K	55K	Return
Collapsed	exp_008d (trained dt_emb)	8.16	−1.86	−4.58	−4.88	∼165
	fixed_dt2 (dt=2)	8.16	−3.02	−5.06	−5.17	∼20
Healthy	exp_008f (no dt_emb)	8.11	−1.85	−4.31	−4.77	∼460
	exp_008a (unexposed)	8.16	−4.88	−5.20	−5.21	∼300
	baseline_s1 (dt=1)	8.10	−4.74	−5.12	−5.18	∼950

Table 28. TCN results across environments and noise magnitudes. Prevention = peak NVE $<$ 2.0× (35K+).

Env	σ	Seed	$\text{NVE}_{\max}$	Prevented?	Eval (%)
Walker	0.1	s0	1.02	✓*	55%
Walker	0.2	s0	0.91	✓	91%
Walker	0.2	s1	0.64	✓	45%
Walker	0.2	s2	2.11†	×	∼5%
Walker	0.2	s3	3.04¶	×	11%
Walker	0.5	s0	3.97	×	27%
Walker	0.5	s1	1.64	✓	41%
Cheetah	0.2	s0	1.90	✓‖	crashed
Cheetah	0.5	s0	10.67	×	26%
Cheetah	1.0	s0	≤0.30	✓§	15%
PickCube	0.2	s0	3.71	×	failing

*Borderline: NVE=1.02× at 95K, below 2.0× threshold but above 1.0×.
‖NVE 1.90× is below the 2.0× prevention threshold but eval crashed; CIVO technically prevented by the NVE criterion despite operational failure.
†NVE = 2.11× at 95K, still rising; eval ∼5% of baseline.
¶Late-onset CIVO: NVE declined to 0.67× at 95K before phase-transition-like spike to 3.04× at 105K; eval 349.6 at 95K → 71.8 at 105K → 104.5 at 215K.
§CIVO prevented (NVE≤0.30); peak eval 116% baseline at 35K, declining to 15% by 215K via progressive policy erosion. See "impossible dose" discussion.

Table 29. DreamerV3 hyperparameters for all experiments. Modifications from defaults are marked with ⋄.

Parameter	Value
World Model	
RSSM discrete states	32 classes × 32 categories
RSSM stochastic dim	32
RSSM deterministic dim	512
MLP hidden size	512
MLP layers	2
Activation	SiLU
Normalization	LayerNorm
Training	
Batch size	16 sequences × 64 steps
Learning rate	3×10^{-4} (Adam, ϵ=10^{-8})
Sequence length	64
Training ratio	512
Free nats	1.0
KL balance (α)	0.8
Discount (γ)	0.997
λ-return	0.95
Policy	
Imagination horizon⋄	8 or 15 (see Table 3)
Actor entropy scale	3×10^{-4}
Actor gradient	Reinforce + dynamics backprop
Temporal Abstraction⋄	
Gate enabled	True (all non-baseline)
Gate fixed dt	1 or 2 (per config)
Subsample max dt	1 (OFF) or 2 (ON)
dt-embedding dim	32
Environment	
Action repeat	2 (Walker Walk, Cheetah Run)
Observation	Proprioceptive (no pixels)
Total environment steps	105K
Evaluation frequency	Every 10K steps
Evaluation episodes	10

Table 30. Retrospective CIVO detector validation (n=14). Detection type: "predictive" = alert precedes 20% eval decline; "reactive" = alert follows decline onset.

Exp	Config	Verdict	Trigger	Type	Peak Eval	r
008d s0	trained dt-emb	TP	45K	predict.	167	0.
010b	trained dt-emb (Cheetah)	TP	65K	reactive	322	3.
008d s3	trained dt-emb	TP	45K	predict.	452	0.
008d s4	trained dt-emb	TP	105K	reactive	383	3.
010b s0	trained dt-emb (Cheetah)	TP	95K	reactive	248	6.
008f s3	no dt-emb	TN	–	–	594	
009b s2	trained dt-emb (self-corr.)	TN	–	–	279	
baseline s1	vanilla	TN	–	–	970	
009d s0	vanilla	TN	–	–	951	
009d s1	vanilla	TN	–	–	920	
019b	trained dt-emb (PickCube)	TN	–	–	26.6	
009c s2	no dt-emb	FP	105K	reactive	359	0.
009a s2	vanilla	FP	85K	predict.	967	0.
009c s3	no dt-emb	TN	–	–	457	

TPR = 5/5 (1.00), FPR = 2/9 (0.22)

Table 31. Dual-critic 2×2 factorial results. Walker Walk unless noted.

Exp	Config	Params	NVE	Eval	Outcome
008d	Single narrow	1.2M	4.12×	82.8	Collapse
033	Wide single	2.4M	4.75×	407.9	CIVO → self-repair
035 s0	Narrow dual (min)	1.2M	2.75×	697.8	CIVO → recovery
039 s1	Narrow dual (min)	1.2M	2.19×	592.7	CIVO → recovery
032 s0	Wide dual (min)	2.4M	≤1.1×	739.7	CIVO prevented
032b s1	Wide dual (min)	2.4M	1.4×	551.8	CIVO prevented
038 s2	Wide dual (min)	2.4M	≤0.3×	120.9*	Non-CIVO collapse
Cross-environment					
034	Cheetah dual	2.4M	1.75×	97.5	Partial recovery (31%)
037	Cheetah wide single	2.4M	3.60×	146.1	CIVO → self-repair
036	PickCube dual	2.4M	2.3×	27.7	Transient CIVO

*CIVO prevented (NVE $\leq 0.3\times$) but non-CIVO eval collapse: eval peaked at 354 (155K), then declined to 120.9 (224K) with actor entropy increasing from -3.59 to -0.45.